

LANCER®

 **TST**
Touchscreen Tower



Operation Manual

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ABOUT THIS MANUAL

This booklet is an integral and essential part of the product. Please carefully read the guidelines and warnings contained herein as they are intended to provide the user with essential information for the continued safe use and maintenance of the product. In addition, it provides **GUIDANCE ONLY** to the user on the correct services and site location of the unit.

The installation and relocation, if necessary, of this product must be carried out by qualified personnel with up-to-date safety and hygiene knowledge and practical experience, in accordance with current regulations.

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BEFORE GETTING STARTED

Each unit is tested under operating conditions and is thoroughly inspected before shipment. At the time of shipment, the carrier accepts responsibility for the unit. Upon receiving the unit, carefully inspect the carton for visible damage. If damage exists, have the carrier note the damage on the freight bill and file a claim with carrier. Responsibility for damage to the dispenser lies with the carrier.

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SAFETY NOTICES

READ ALL SAFETY INSTRUCTIONS BEFORE USING THIS UNIT.

This manual contains important safety information and all applicable safety precautions must be observed. To reduce the risk of fire, electric shock, damage to the equipment or personal injury when using this unit all instructions/warnings on the product being used must be followed:

! WARNING

Text following the Warning signal indicates a hazardous situation, which if not avoided, will result in death or serious injury. Be sure to read all Warning statements before proceeding with the installation.

! CAUTION

Text following the Caution signal indicates a hazardous situation, which if not avoided, could result in death or serious injury. Be sure to read the Caution statements before proceeding with the installation

! ATTENTION

Text following the Attention signal addresses a situation that if not followed could potentially damage the equipment. Be sure to read the Attention statements before proceeding

NOTE

Text following the Note signal provides you with information that may help you more effectively perform the installation procedures within this manual. Disregarding information will not cause damage or injury, however it may limit the performance of the dispenser.

SAFETY INSTRUCTIONS

! Intended Use

- The dispenser is for indoor use only
- This appliance is intended to be used in commercial applications such as restaurants or similar.
- This appliance is to be installed in a location where its use can be overseen by trained personnel.
- This appliance should not be used by children or infirm persons without supervision.
- This appliance is not intended for use by persons (including children) with reduced physical, sensory or mental capabilities, or lack of experience and knowledge, unless they have been given supervision or instruction concerning use of the appliance by a person responsible for their safety.
- This appliance can be used by children aged from 8 years and above and persons with reduced physical, sensory or mental capabilities or lack of experience and knowledge if they have been given supervision or instruction concerning use of the appliance in a safe way and understand the hazards involved.
- Cleaning and user maintenance shall not be performed by children without supervision.
- This unit is not a toy and children should be advised not to play with the appliance.
- The min/max ambient operating temperature for the dispenser is 40°F to 90°F (4°C to 32°C).
- Do not operate unit below minimum ambient operation conditions.
- Should freezing occur, cease operation of the unit and contact authorized service technician.
- The maximum tilt for safe operation is 5°.
- This appliance must be installed and serviced by a professional.

5 Carbon Dioxide (CO₂)

- **WARNING:** Carbon Dioxide (CO₂) is a colorless, noncombustible gas with a light pungent odor. High percentages of CO₂ may displace oxygen in the blood.
 - **WARNING:** Prolonged exposure to CO₂ can be harmful. Personnel exposed to high concentrations of CO₂ gas will experience tremors which are followed by a loss of consciousness and suffocation.
- **WARNING:** If a CO₂ gas leak is suspected, immediately ventilate the contaminated area before attempting to repair the leak.
- **WARNING:** Strict attention must be observed in the prevention of CO₂ gas leaks in the entire CO₂ and soft drink system.

F Power

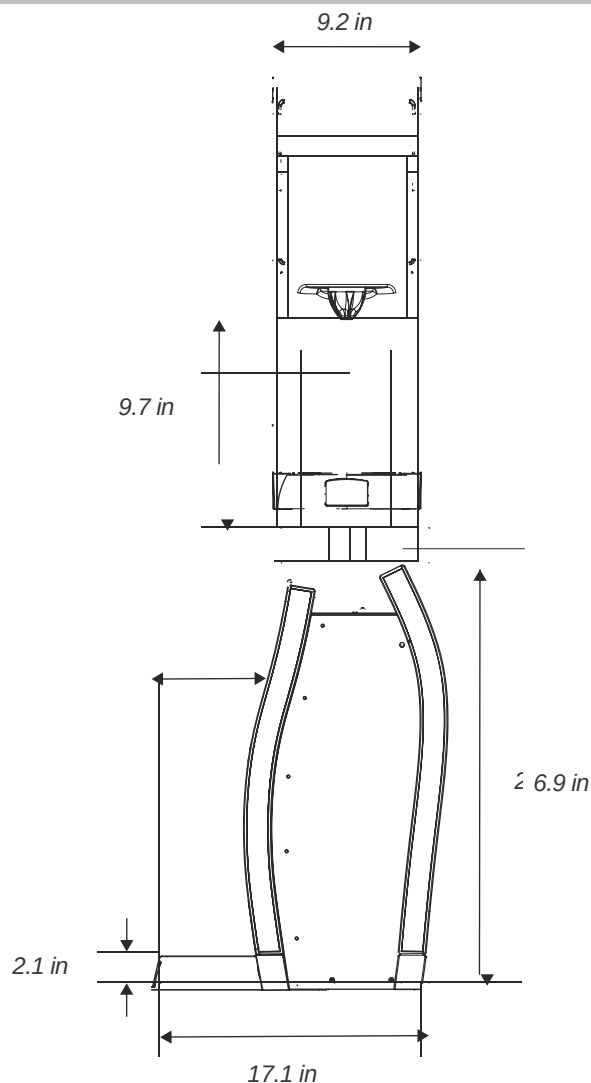
- Follow all local electrical codes when making connections.
- Appliance must be supplied by 24 VDC.
- Check dispenser name plate label, located inside tower panels, for correct electrical requirements of unit. **DO NOT** plug into a wall electrical outlet unless the current shown on the serial number plate agrees with local current available.
- Each dispenser must have a separate electrical circuit.
- **DO NOT** use extension cords with this unit.
- **DO NOT** 'gang' together with other electrical devices on the same outlet.
- **WARNING:** Always disconnect electrical power to the unit to prevent personal injury before attempting any internal maintenance.
 - The resettable breaker switch should not be used as a substitute for unplugging the dispenser from the power source to service the unit.

! Water Notice

- Appliance is not suitable for installation where a water jet could be used.
- Provide an adequate, potable water supply. Water pipe connections and fixtures directly connected to a potable water supply must be sized, installed, and maintained according to federal, state, and local codes.
 - The water supply line must be at least a 9.525 mm (3/8 inches) pipe with a minimum of 20 psi (0.137 MPa) line pressure, but not exceeding a maximum of 65 PSI (0.448 MPa). Water pressure exceeding 65 PSI (0.448 MPa) must be reduced to 65 PSI (0.448 MPa).
 - Use a filter in the water line to avoid equipment damage and beverage off-taste. Check the water filter periodically, as required by local conditions.
- **CAUTION:** The water supply must be protected by means of an air gap, a back flow prevention device (located upstream of the CO₂ injection system) or another approved method to comply with NSF standards. A leaking inlet water check valve will allow carbonated water to flow back through the pump when it is shut off and contaminate the water supply.

Ennen asennusta

Tekniset tiedot



MITAT

Leveys: 235 mm
Syvyys: 433 mm
Korkeus: 682 mm

PAINO

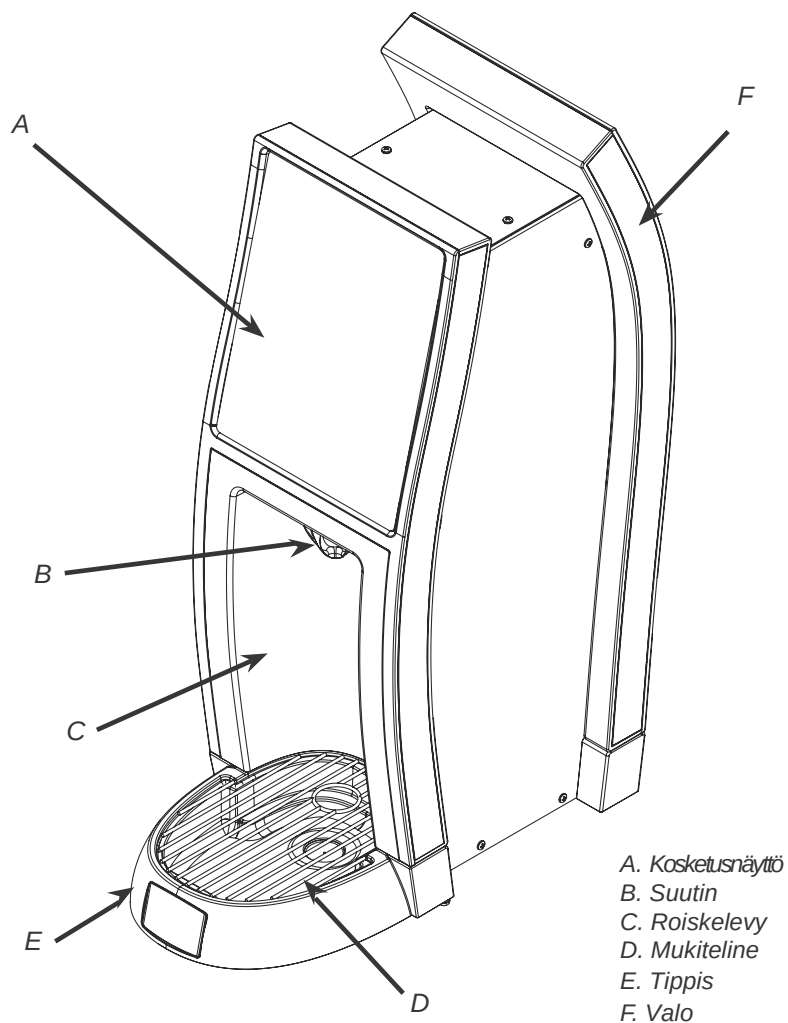
Pakkauksessa: 18.14 kg
Käytössä: 15.88 kg

Jännite / Virta

24 VDC / 2.0 Amps

Virtaus

45-60ml sekunnissa



Vedensyöttö

Min : 1,4bar
Max : 4,5bar

Co2 syöttö

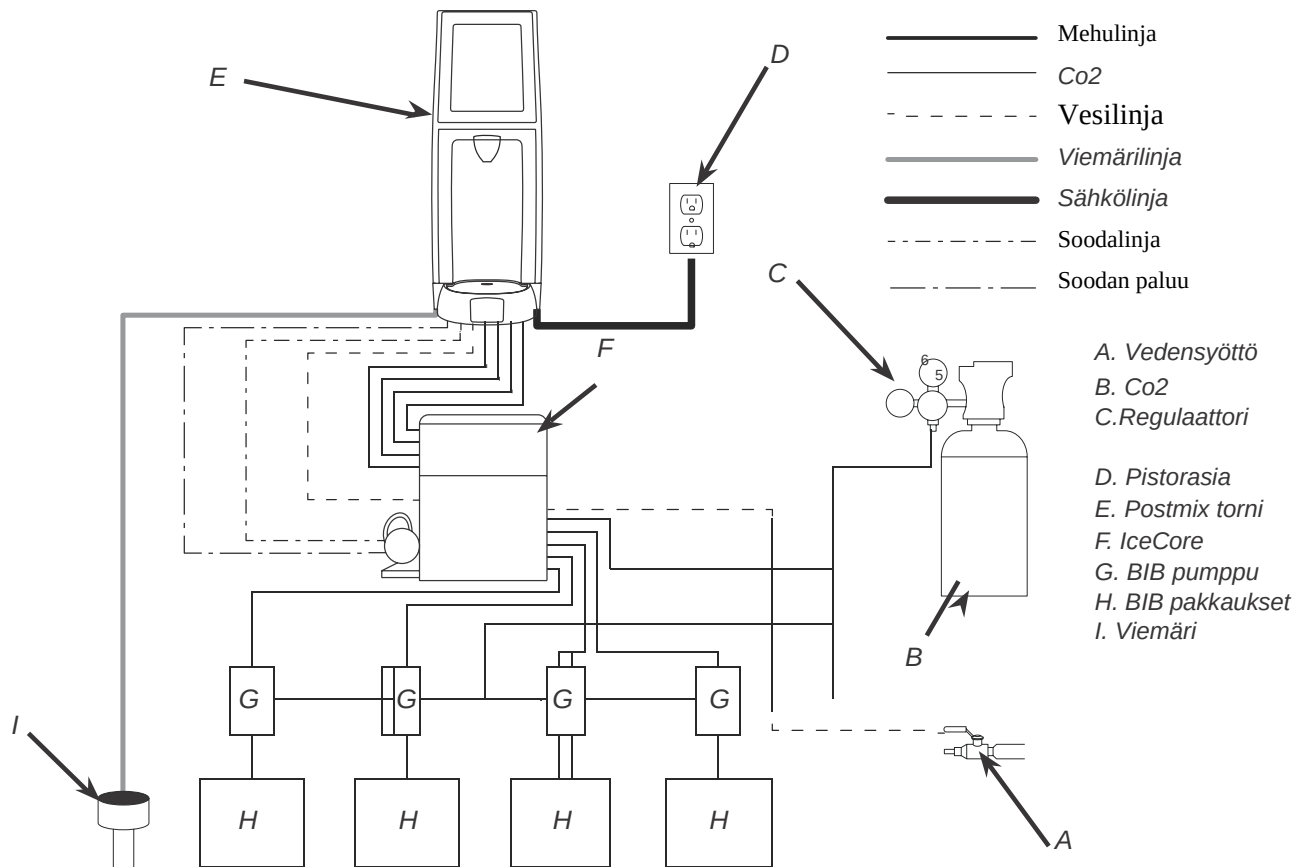
Min : 4,8bar
Max : 5,5bar

Liittimet

Soodan sisään: 3/8 inch (9.5 mm) barb
Sooda ulos: 3/8 inch (9.5 mm) barb
Vesi sisään: 3/8 inch (9.5 mm) barb
Siirapit sisään: 3/8 inch (9.5 mm) barb
Viemäri: 5/8 inch (15.9 mm) barb

Laitteen äänenvoimakkuus on alle 70db

Yleiskatsaus



Asennukseen tarvittavat tavarat

Työkalut:

- ☐ Klemmaripihdit
- ☐ Letkunkatkaisu pihdit
- ☐ Jakoavain
- ☐ Talttapäämeisseli
- ☐ Ristipäämeisseli
- ☐ Akkuporakone

POSTMIX tavarat :

- ☐ CO₂ Regulaattori
- ☐ CO₂ Pullo
- ☐ Kahle
- ☐ Postmix torni
- ☐ Mehuletku
- ☐ Klemmareita

Huomioi seuraavat ennen

asennusta:

- ☐ Vesipisteen sijainti
- ☐ Viemärin sijainti
- ☐ Pistorasia
- ☐ Ilmanvaihdon riittävyys

BIB tavarat:

- ☐ BIB Rack
- ☐ BIB uutelaatikot
- ☐ BIB Pumput
- ☐ BIB Liittimet

INSTALLATION

Read This Manual

This manual was developed by Lancer Corporation as a reference guide for the owner/operator and installer of this dispenser. Please read this manual before installation and operation of this dispenser. Please see pages 23 - 25 for troubleshooting or service assistance. If the service cannot be corrected please call your Service Agent or Lancer Customer Service. Always have your model and serial number available when you call.

Unpacking the Dispenser

1. Cut package banding straps and remove.
2. Open the box and remove the accessory kit and loose parts.
3. Carefully lift the unit out of the box and place on a flat surface taking care to not scratch the plastic covers.

NOTE

Inspect unit for concealed damage. If evident, notify delivering carrier and file a claim against the same.

Selecting/Preparing a Counter Location

NOTE

The dispenser should only be installed in a location where it can be overseen by trained personnel

1. Select a location that is in close proximity to a properly grounded electrical outlet, within five (5) feet (1.5 m) of a drain, and a water supply that meets the requirements shown in the Specifications section found on page 4.

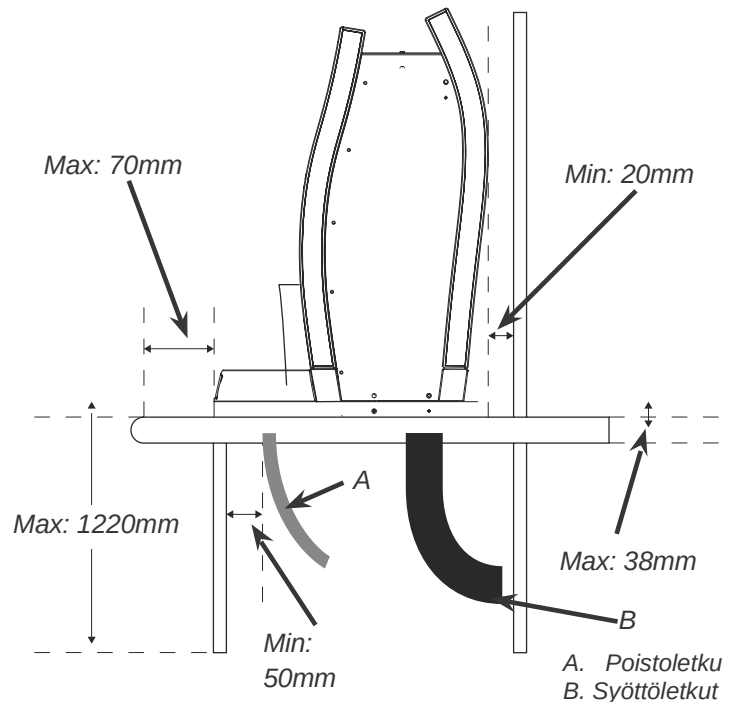
! ATTENTION

Inspect the counter location where the unit is to be installed. Verify the selected counter is strong enough to safely support the weight of the installed unit, after the cutout for the unit is made. The ideal counter for installation should measure at least one (1) inch thick.

2. Select a location that utilizes the clearances/space required for installation.
3. Select a location for the remote chiller system or carbonator (if necessary), syrup pumps, CO₂ tank, syrup containers, and water filter (recommended).
4. Using Counter Cutout Template provided (See page 31), cut out required opening for the tower installation in the designated location.

NOTE

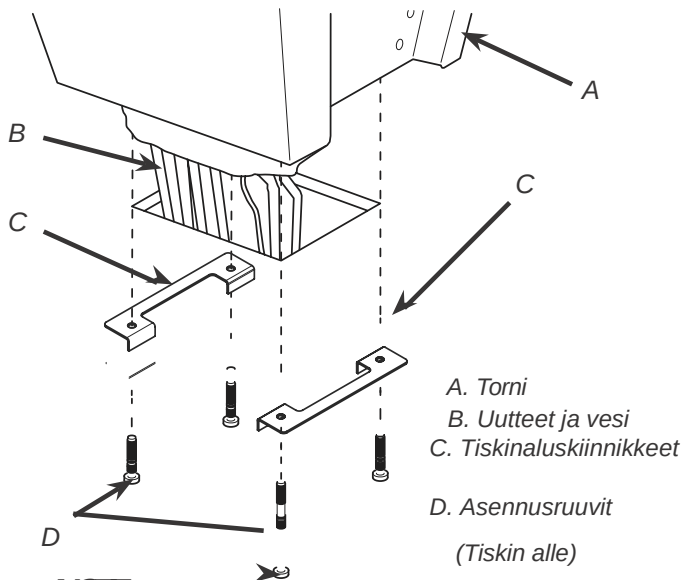
To assure that beverage service is accessible to all customers, Lancer recommends that counter height and equipment selection be planned carefully. The 2010 ADA Standards for Accessible Design states that the maximum reach height from the floor should be no more than 48" if touch point is less than 10" from the front of the counter, or a maximum of 46" if the touch point is more than 10" and less than 27" from the front of the counter. For more information about the customer's legal requirements for the accessibility of installed equipment, refer to 2010 ADA Standards for Accessible Design - <http://www.ada.gov>.



Asennus

1. Using proper lifting techniques, lift the tower over the counter top, then route end of power cord through the designated opening in the counter.
2. Slide the tower into the cutout in the counter and use the screws and mounting brackets provided to secure the tower to the counter.

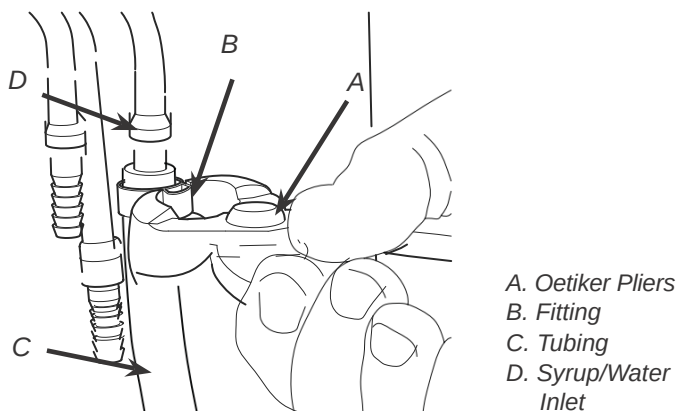
KUN ASENNAT TORNIA TISKIIN, VARMISTA ETTEI RUUVIT TULE YLI 19MM YLI ASENNUSLEVYN. TÄMÄ SAATTAA VAHINGOITTA VENTTIILEITÄ!!!



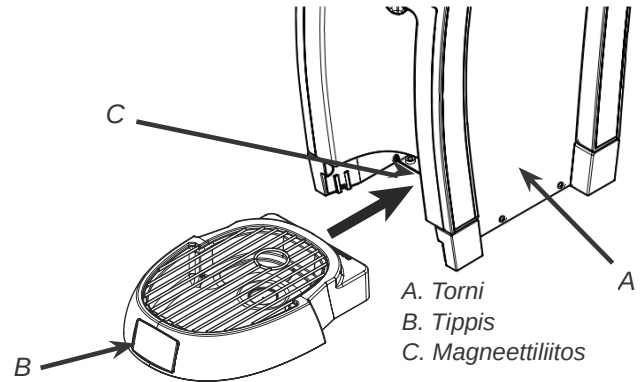
NOTE

NSF listed units must be sealed to the counter.

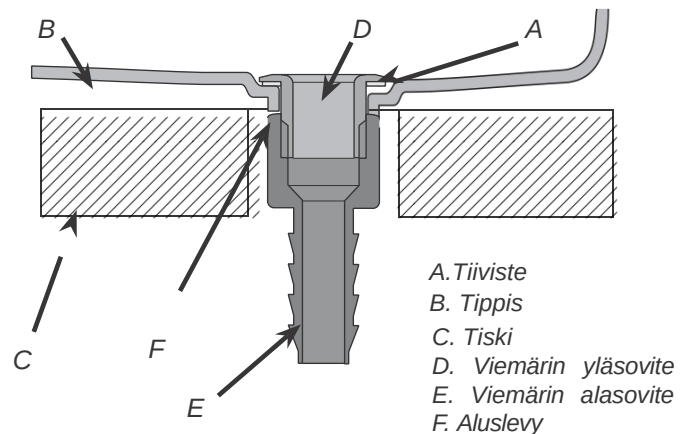
3. Route appropriate tubing from the syrup pump location to the syrup inlets on the tower. Connect tubing to inlets using the oetiker pliers and fittings. Repeat for all syrup connections.
4. Route appropriate tubing for the carbonated/plain water inlets then connect tubing to inlet. Repeat for all water connections including the return inlet.



5. Slide drip tray into opening in tower until magnets engage and lock drip tray in place. Connect ADA wire harness from drip tray to ADA wire harness in tower.



6. If a drain line is to be utilized, place gasket at the bottom of the drain then attach drain fitting to drip tray.



! ATTENTION

If a drain line is utilized, then connect ADA wire harness in drip tray to wire harness behind splash plate before attaching drip tray.

7. Plug in power cord to power supply then route power supply cord to the designated grounded electrical outlet.

! WARNING

DO NOT PLUG UNIT INTO GROUNDED ELECTRICAL OUTLET AT THIS TIME. Make sure that all water lines are tight and unit is dry before making any electrical connections

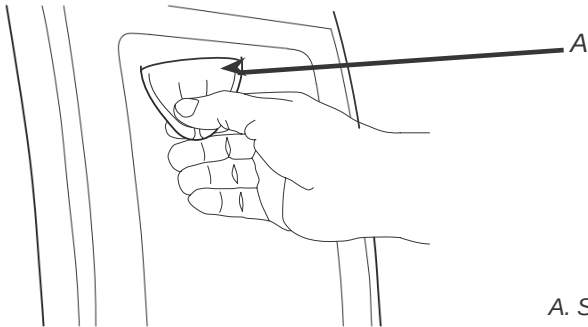
NOTE

Unit has the option to be supported by a remote chiller system or remote ice cooled system. Please see the manufacturer's specifications and instructions for installation. The following are the instructions for plumbing the remote chiller system to the tower.

8. Route appropriate tubing from the syrup pump location to the syrup inlets at the remote chiller. Repeat for all syrup connections.
9. Route appropriate tubing from the water source to the water inlet at the remote chiller and only connect tubing to the water source.
10. Turn on the water and flush the water line thoroughly.
11. Turn off the water and connect water line to the plain water

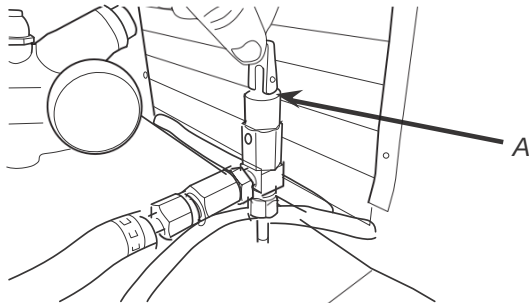
Ohjelmointi

1. Poista mukiteline tippiksestä
2. Poista suutin.



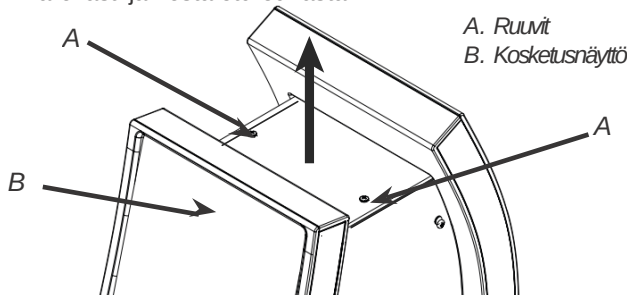
A. Suutin

3. Poista roiskelevy
4. Kytke vesi päälle
5. Tarkista, ettei BIB- liittimissä ole vuotoja
6. Avaa paineenvapautusventtiili joka sijaitsee IceCoren luona kääntämällä vipua ylös. Pidä auki kunnes vesi tulee läpi, sen jälkeen sulje venttiili.



A. Paineenvapautusventtiili

7. Avaa kaksi ruuvia päältä (A). Sen jälkeen liu'uta levyä taakse varovasti ja nosta etureunasta.



A. Ruuvit
B. Kosketusnäyttö

8. Yhdistä laite verkkovirtaan

! WARNING

The dispenser must be properly electrically grounded to avoid serious injury or fatal electrical shock. The power cord has a three-prong grounded plug. If a three-hole grounded electrical outlet is not available, use an approved method to ground the unit. Follow all local electrical codes when making connections. Each dispenser must have a separate electrical circuit. Do not use extension cords. Do not connect multiple electrical devices on the same outlet.

9. Kytke laite päälle. Kytin löytyy päällilevyn takaa joka irrotettiin kohdassa 7.
10. Kun kone on käynnistynyt, mene huoltovalikkoon painamalla sormella ruudun oikeaa yläkulmaa.
11. Liu'uta sormi vasempaan yläkulmaan ja pidä sitä paikallaan vähintään kaksi sekuntia.



- Liu'uta sormi vasemmalle

12. Kun olet pitänyt sormi paikallaan vähintään kaksi sekuntia, paina kaikkia neljää nurkkaa .



13. Näppäimistö tulee ruudulle

NOTE

Contact Lancer Technical Support for the units' designated pin number.

14. Kaikkiin asetuksiin päästäksesi, paina oikeaa yläkulmaa vähintään viisi (5) sekuntia ja syötä koodi (6655).

NOTE

The manager's access to the service menu allows access to the Lighting screen (see page 15), the Sold Out screen (See page 16), and the Time & Delays screen (See page 14).

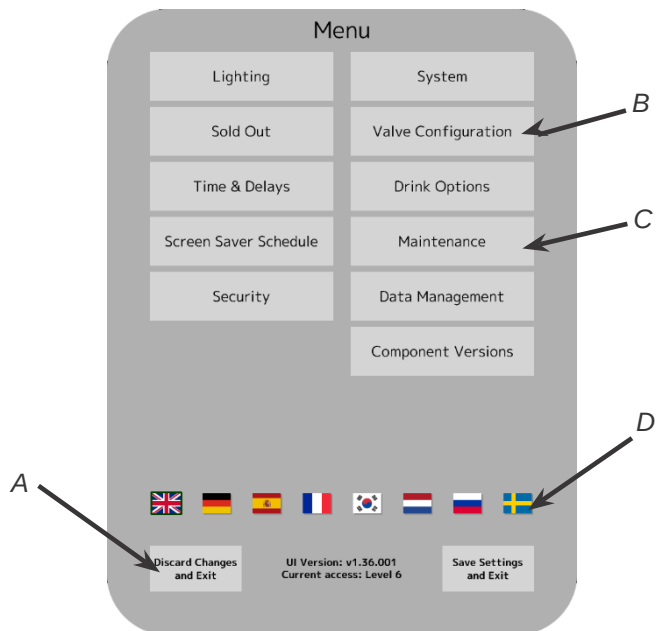
15. Laitteen lukitseminen tai laitto "lepotilaan": Syötä koodi (3.14).

NOTE

Tämä tila estää asiakasta laskemasta juomaa ja säästää virtaa kun laite ei ole käytössä.

16. Pääsy "Loppuunmyyty" valikkoon, syötä koodi (963.).

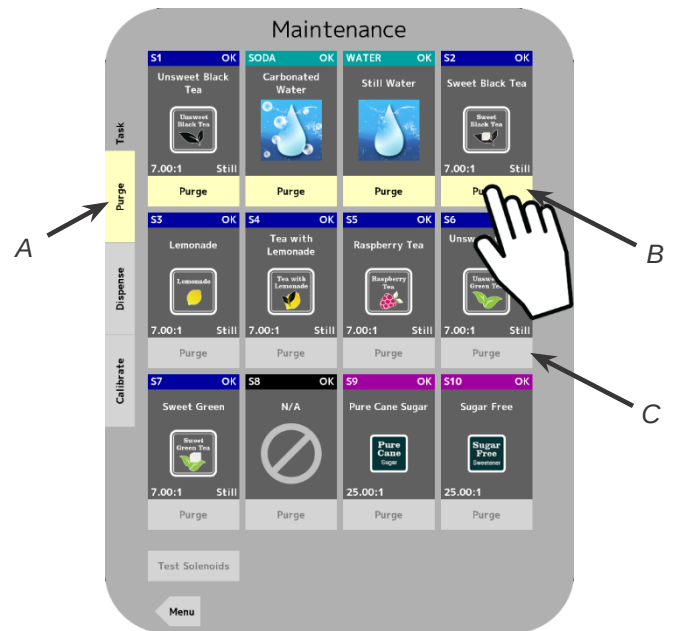
17. Valikossa paina Maintenance nappia.



A. Hylkää muutokset
B. Venttiilien konfigurointi
C. Maintenance nappi
D. Sijainnin muutos

18. Paina purge (puhdistus) ruudun vasemmassa reunassa.

19. Paina purge sekä vedelle, että soodalle.



A. Purge nappi
B. Max 4 purge nappia voi olla valittuna
C. Harmaa Purge nappi

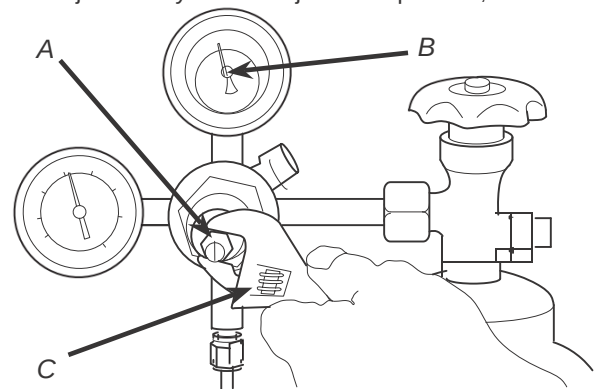
NOTE

Kun purge on aktivoitu, laite laskee tuotetta kunnes se deaktivoidaan. Purgen deaktivoinniseksi, paina purge-nappia uudestaan. Maksimissaan neljää linjaa voidaan puhdistaa samaan aikaan. Kun neljä linjaa on valittuna, kaikki muut muuttuvat harmaiksi eikä niitä voi valita.

20. Kun vesi virtaa nästisti, paina purgea uudestaan sammuttaaksesi syötön.

21. Varmista, että hapotin ei ole päällä ennenkuin kytket hiilarin.

22. Sen jälkeen kytke hiilari ja säädä paine 4,5bariin.



A. Regulaattorin säätö
B. Säädä 4,5bariin
C. Jakoavain

23. Puhdista soodalinjaa niin kauan, että kaasu loppuu.

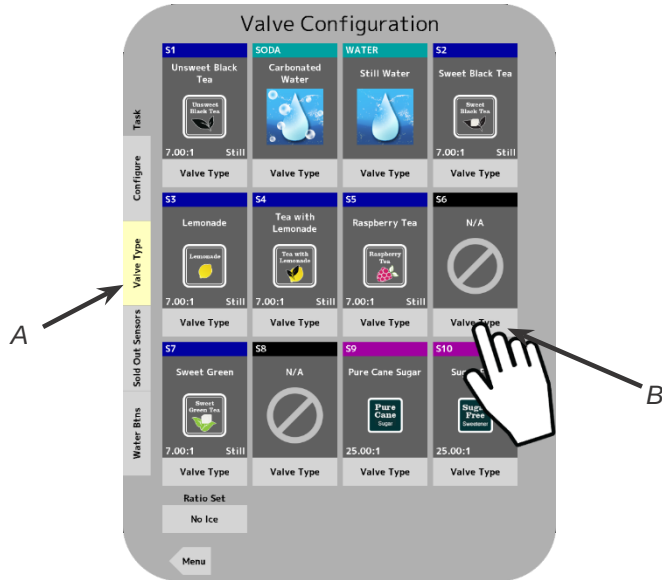
24. Käynnistä hapotin.

25. Laske soodaa uudestaan niin kauan, että se hapottuu.

26. Laske tuotetta jokaisesta uutelinjasta läpi niin kauan, että se virtaa nästisti.

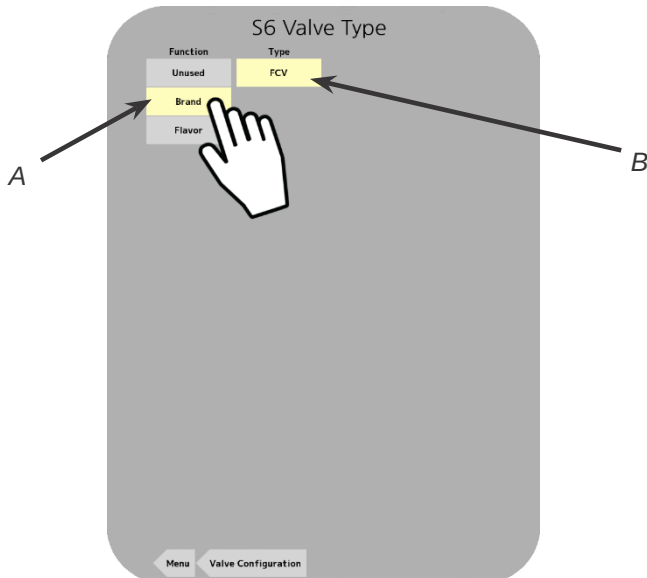
Maun lisääminen

1. Uuden tuotemerkin lisäämiseksi moduuli on ensin aktivoitava.
2. Paina Huolto-valikosta *Valve Configuration* nappia.
3. Paina *Valve Type* ruudun vasemmasta reunasta, sen jälkeen valitse joku ei-aktiivinen moduuli.



A. Valve Type
B. Valve Type

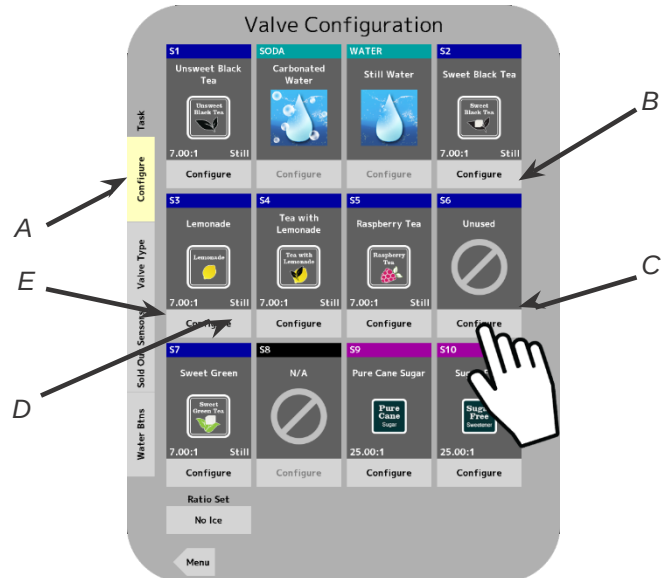
4. Valitse haluttu toiminto ja venttiilin tyyppi valittuun venttiilimoduuliin.



A. Venttiilin toiminto
B. Venttiilin tyyppi

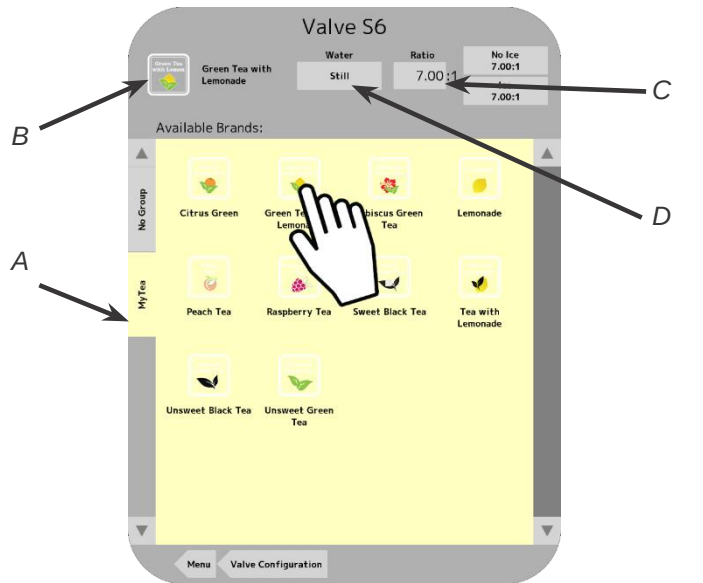
5. Paina *Valve Configuration* nappia palataksesi Valve Configuration valikkoon.
6. Toista kohdat 3 ja 4 kaikkiin haluamiisi moduuleihin.
7. Valve Configuration valikossa, paina *Configure* ruudun vasemmassa reunassa.

8. Paina *Configure* nappia minkätahansa aktivoitun brändin tai makumoduulin kohdalla avataksesi sen asetukset.



A. Configure väli
B. Configure nappi
C. Aktivoitu moduuli
D. Veden tyyppi
E. Suhde

9. Valitse uusi brändi kirjastosta joka on ruudun vasemmassa reunassa.



A. Kirjasto
B. Asetettava logo
C. Suhde
D. Veden tyyppi

NOTE

Each brand has a default water type and ratio already set when they are selected. The water type and ratio can be adjusted if necessary, however, adjusting the ratio here is purely representational and does not set the ratio for the finished drink. Adjust the ratio by selecting the Ice or No Ice ratio using the buttons in the upper right corner of the screen, or by manually adjusting the ratio by tapping the number and entering the new value on the keypad.

10. Kun brändi / maku on asetettu moduuliin, paina Valve configuration nappia palataksesi Valve Configuration ruutuun.
11. Toista kohdat 8 ja 9 kaikkiin haluamiisi tuotteisiin.
12. Paina Menu nappia palataksesi alkuvalikkoon.

13. Service Menussa, paina Maintenance nappia.
14. Paina Purge välilehteä vasemmassa reunassa.
15. Laske uusia tuotteita niin kauan, että virtaa nättisti (Sivu 9)
16. Paina Menu nappia palataksesi päävalikkoon.

CALIBRATION & MAINTENANCE

Veden kalibrointi

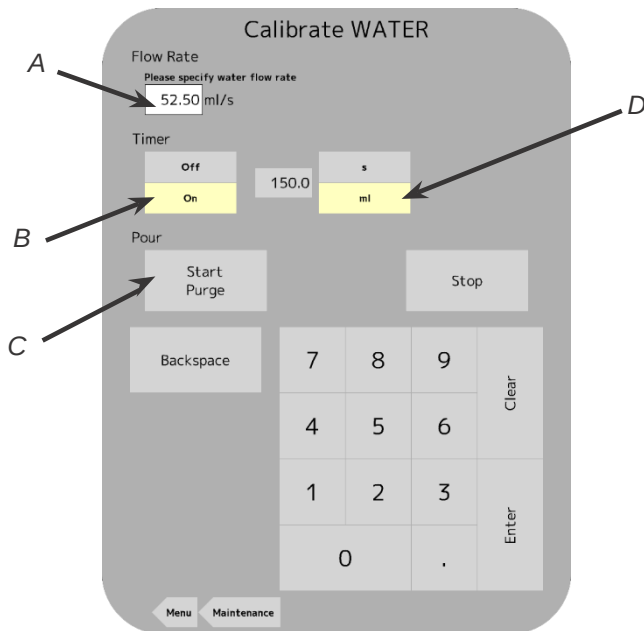
1. Asenna suutin. (Sivu 8)
 2. Päävalikossa, paina Maintenance.
 3. Paina Calibrate välilehteä ruudun vasemmassa laidassa ja paina Calibrate nappia vesimoduulille.
 4. Syötä haluttu virtaus millilitroina sekunnissa (ml/sec).
- Tämä säätää valmiin juoman virtauksella 60ml/sec.

ESIMERKKI

Valmiin juoman virtaus = 60 ml/sec
Suhde = 7:1

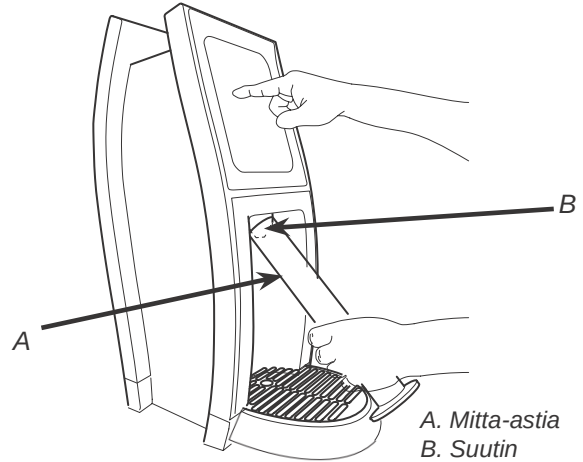
$60 \text{ ml/s} \times 7 / (7 + 1) = 52.5 \text{ ml/sec}$ veden virtaus

$60 \text{ ml/s} \times 1 / (7 + 1) = 7.5 \text{ ml/sec}$ uutteen virtaus

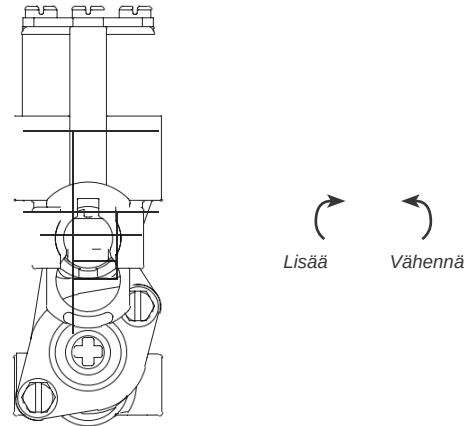


5. Aseta Timer ON asentoon ja valitse millilitrat (ml) mittayksiköiksi.
6. Syötä näppäimistöllä tietty annosteltava tilavuus perustuen kalibrointiin käytettävän mitta-astian kokoon. Mitä suurempi annosteltu tilavuus, sitä tarkemmat tulokset. Käytä esimerkiksi 150 ml.

7. Aseta mitta-astia suuttimen alle ja paina purge. Laite annostelee edellisessä vaiheessa asetetun mitan.



8. Tarkista mitta.
Jos mitta ei täsmää asetettuun arvoon (150 ml) joka syötettiin aiemmin, poista venttiilin suoja ja säädä meisselillä soodan virtausta. (Sivulla 29 venttiilikartta).



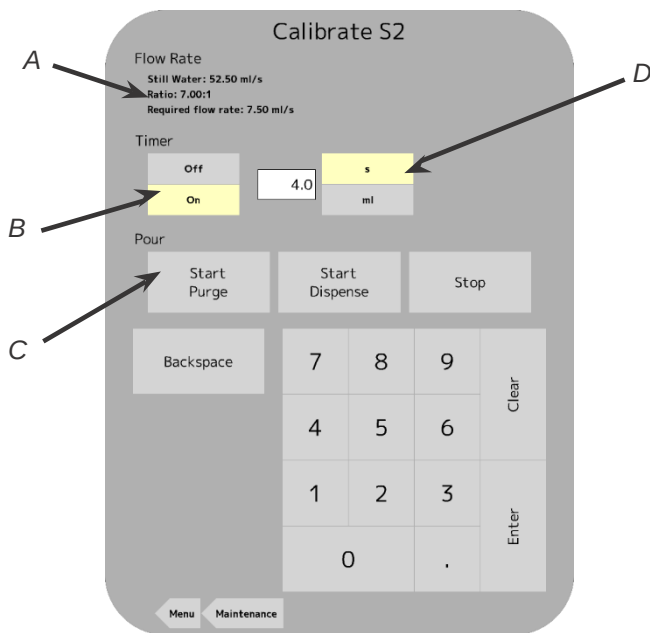
9. Toista kohdat 6 ja 7 kunnes haluttu 150 ml on saavutettu.
10. Toista kohdat 2-8 soodalle.

Makujen kalibrointi

NOTE

JÄÄHDYTTIMEN TULISI KÄYDÄ VÄHINTÄÄN TUNTI ENNEN BRIXAUSTA.
JUOMAN LÄMPÖTILAN TULISI OLLA MAKSIMISSAAN **4.4°C**
PARAS OLISI JOS JÄÄHDYTTÄJÄSSÄ OLISI JO JÄÄPANKKI.

1. Service menussa, paina Maintenance nappia.
2. Paina calibrate välilehteä ruudun vasemmassa reunassa ja paina Calibrate nappia ensimmäiselle siirappimoduulille .
3. Veden virtaus pitäisi olla jo säädettyinä jääveden kalibroinnin jäljiltä ja suhde valmiina tuotteen konfiguroinnista (Sivu 10)



A. Virtaus / suhde
B. Ajastin
C. Laskunappi
D. Yksikkö

NOTE

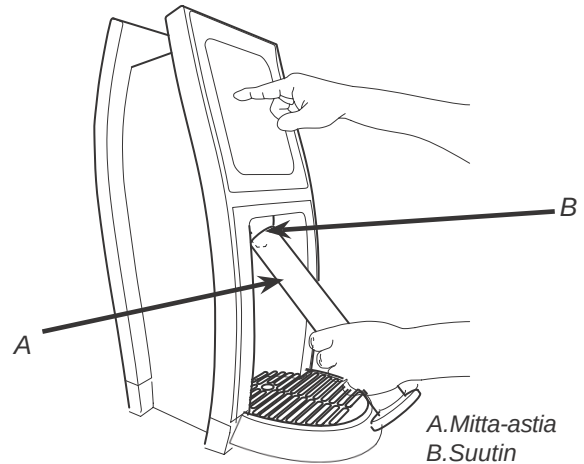
Uutemoduulit voidaan kalibroida samallatavalla kuin vesi. Käyttämällä tilavuutta (ml). Alla on ohjeet ajalla tilavuuden sijasta.

4. Sääda timer ON asentoon ja valitse (s) mittayksiköiksi.
5. Sääda näppäimistöllä laskuajaksi 4 sekuntia..

NOTE

Valmiin juoman virtaukseksi on säädetty 60 ml/sec, tämä tekee uutteen virtaukseksi 7.5 ml/s. 4 sekunnissa uutetta pitäisi tulla 30 ml.

6. Aseta mitta-astia suuttimen alapuolelle, paina Start Purge nappia. Laite annostelee uutetta 4 sekuntia.



7. Tarkista laskettu mitta.
Jos mitta ei vastaa haluttua 30ml (taulukko seuraavalla sivulla) poista venttiilin suoja ja säädä manuaalisesti.
8. Toista kohdat 6 ja 7 kunnes haluttu 30 ml on saavutettu.
9. Toista kohdat 2-8 loppuihin venttiileihin.

B

Taulukko näyttää virtauksen vesimoduuleille ja uutteen määrän neljän sekunnin laskulla eri suhteilla (Kaikki valmista juomaa 60 ml/sec):

Suhde	5	5.2	5.4	5.6	5.8	6	6.2	6.4	6.6	6.8	7
Vesi (ml/s)	50	50.32	50.63	50.91	51.18	51.43	51.67	51.89	52.11	52.31	52.50
Uutteen määrä (4 sec.) (ml)	40	38.71	37.50	36.36	35.29	34.29	33.33	32.43	31.58	30.77	30

Suhde	7.2	7.4	7.5	7.6	7.8	8	8.2	8.4	8.6	8.8	9
Vesi (ml/s)	52.68	52.86	52.94	53.02	53.18	53.33	53.48	53.62	53.75	53.88	54
Uutteen määrä (4 sec.)	29.27	28.57	28.24	27.91	27.27	26.67	26.09	25.53	25	24.49	24

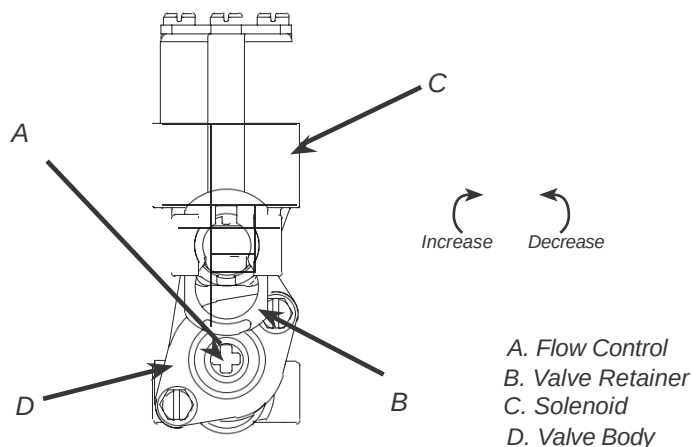
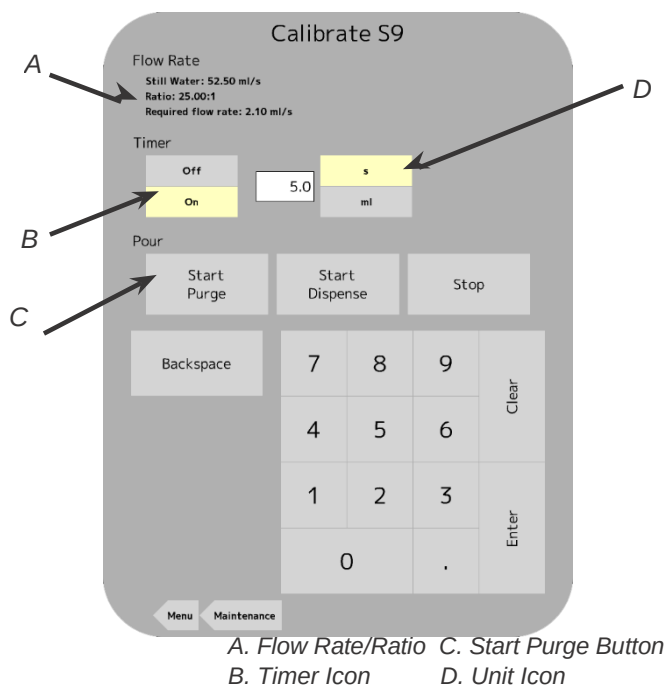
Calibrating Flavor Modules (Shotit / ei käytössä)

1. From the Maintenance menu, press the Calibrate tab on the far left side of the screen and press the Calibrate button for any designated flavor module.
2. The water flow rate should be set from the calibration of the plain water module in the previous section and the ratio should be determined from when the flavor was configured. (See page 10, *Adding New Brand/Flavor Module*)

NOTE

The finished drink flow rate was set to 60 ml/sec, which makes the finished flavor flow rate 2.1 ml/s. In 5 seconds, the volume of syrup that should be dispensed is 10.5 ml.

5. With the graduated cylinder placed in a position below the nozzle, press the Start Purge button. The unit will dispense the designated flavor for 5 seconds.
6. Examine the dispensed volume in the graduated cylinder. If the dispensed volume does not match the value of 10.5 ml, (see note above) remove the protective cap for the corresponding valve and use a screwdriver to adjust the flavor flow control. (See Plumbing Diagram on page 29 for reference).



3. Set the Timer to the ON position and select seconds (s) as the desired unit of measurement.
4. Using the keypad, enter in a time of 4 seconds as the preset dispensing time.

7. Repeat steps 6 and 7 if any more bonus flavor flow adjustment is necessary.
8. Repeat steps 2-8 for any remaining bonus flavor module.
9. Re-install top plate, splash plate, and cup rest. (See page 8)

Scheduled Maintenance

As Needed	Keep exterior surfaces of dispenser (include drip tray and cup rest) clean using a clean, damp cloth.
Daily	- Remove outer nozzle and rinse well in warm water. DO NOT use soap or detergent. This will cause foaming and off taste in finished product. - Using a soft cloth and cleaning solution, clean the nozzle injectors. See Cleaning and Sanitizing Nozzle section on page 21 for reference. - Remove cup rest and wash in cleaning solution. - Pour warm cleaning solution into the drip tray and wipe with a clean cloth. - With a clean cloth and cleaning solution, wipe off all of the unit's exterior surfaces and splash areas. DO NOT USE ABRASIVE SOAPS OR STRONG DETERGENTS. DO NOT USE AMMONIA BASED PRODUCTS WHEN CLEANING THE SCREEN OR SURROUNDING PLASTICS. - Replace the cup rest and nozzle.
Weekly	Taste each product for off tastes. If off taste occurs clean and sanitize the unit using the
	appropriate procedures outlined in the Cleaning and Sanitizing section of this manual.
	Sanitizing section of this manual.
Every Six Months	- Clean remote chiller according to manufacturer's instructions (if necessary). - Clean the entire exterior of the unit.

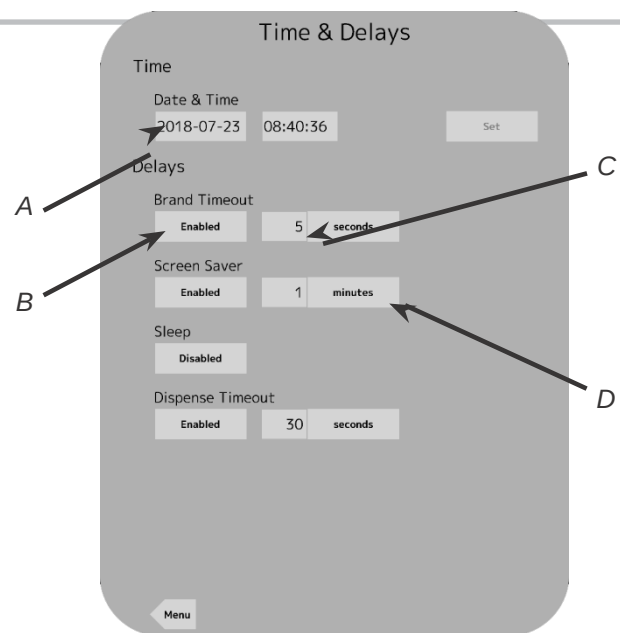
Ominaisuudet

Aikaja ajastukset

1. Service Menussa, paina *Time & Delays* nappia päästäksesi Time & Delays valikkoon.
2. Päivitä aika ja päivämäärä tarvittaessa. Paina *Set* tallentaaksesi.
3. Enable (päälle) Disable (pois)

NOTE

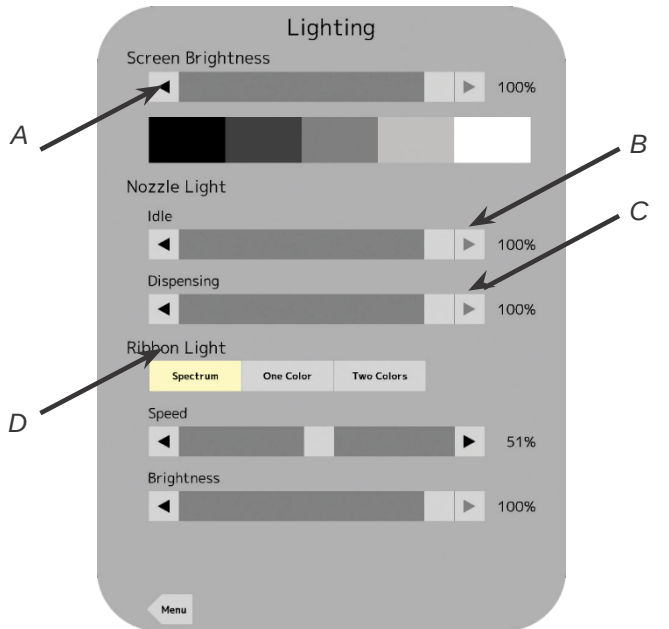
Brand Timeout – Kuinka kauan brändi näkyy ruudulla
Screen Saver – Näytönsäästäjän ajastus
Sleep – Aika jonka jälkeen laite menee lepotilaan (sivu 4)
Dispense Timeout – Tuotteen laskun aikakatkaisu



- A. Päivämäärä&aika
 B. Päälle / pois
 C. Aika
 D. Yksikkö

Valaistus

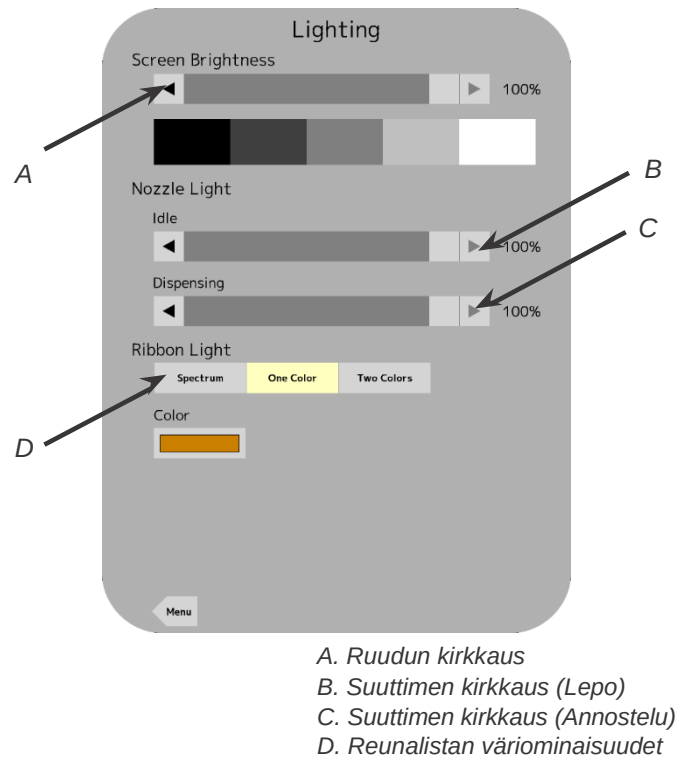
1. Service menussa, paina *Lighting* nappia.
2. Täällä voit säätää ruudun kirkkautta, suuttimen kirkkautta kun laite annostelee juomaa ja kun ei annostele. Lisäksi voit säätää laitteen reunavaloja.



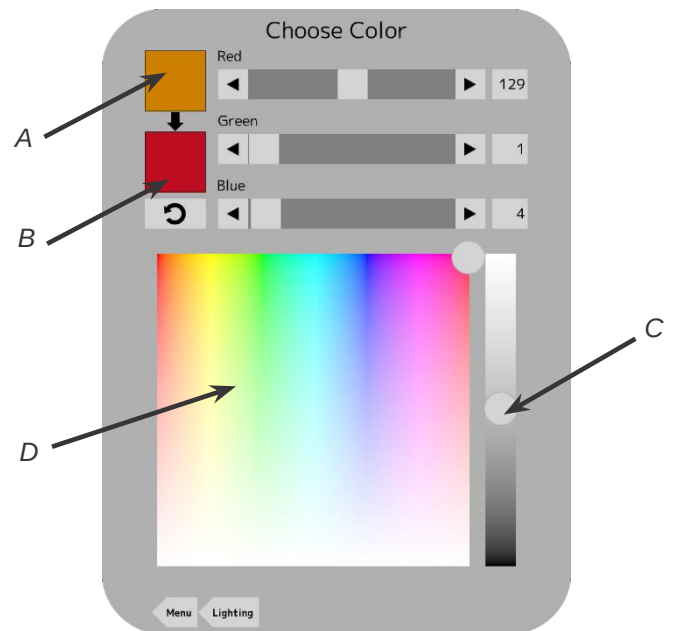
- A. Ruudun kirkkaus
 B. Suuttimen kirkkaus (Lepo)
 C. suuttimen kirkkaus (Annostelu)
 D. Reunavalaistus

NOTE

Spectrum – Vaihtuvat valittujen värien mukaan. Kirkkaus ja nopeus voidaan valita liukukytkimillä.
 One Color – Vain yksi väri.
 Two Colors – Kaksi väriä



- A. Ruudun kirkkaus
 B. Suuttimen kirkkaus (Lepo)
 C. Suuttimen kirkkaus (Annostelu)
 D. Reunalistan väriominaisuudet



- A. Vanha väri
 B. Uusi väri
 C. Kirkkaus
 D. Värisävy

Sold-Out Feature

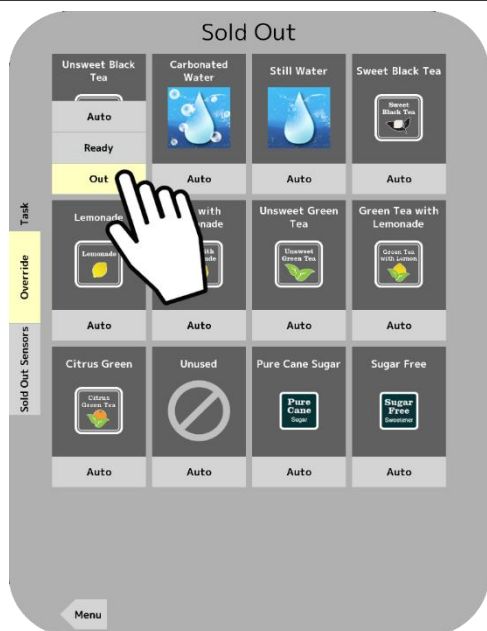
1. From the Service Menu, press the *Sold Out* button.
2. Press the *Override* tab on the far left side of the screen.
3. From here, manually adjust specific brands to read *Ready*, *Out*, or *Auto*

NOTE

Ready - signifies there is available product and the valve will dispense when activated

Out - signifies there is no available product or there is a problem with the specified brand and will dispense when activated.

Auto - signifies that the configured Sold Out Sensor controls whether the brand can be dispensed. This feature requires an optional sold out sensor kit, does not come standard, and is available for up to ten (10) brands at one time. The following is a set of instructions on how to set up this feature. If no sold out sensor is assigned then the Auto feature acts the same as the Ready feature.

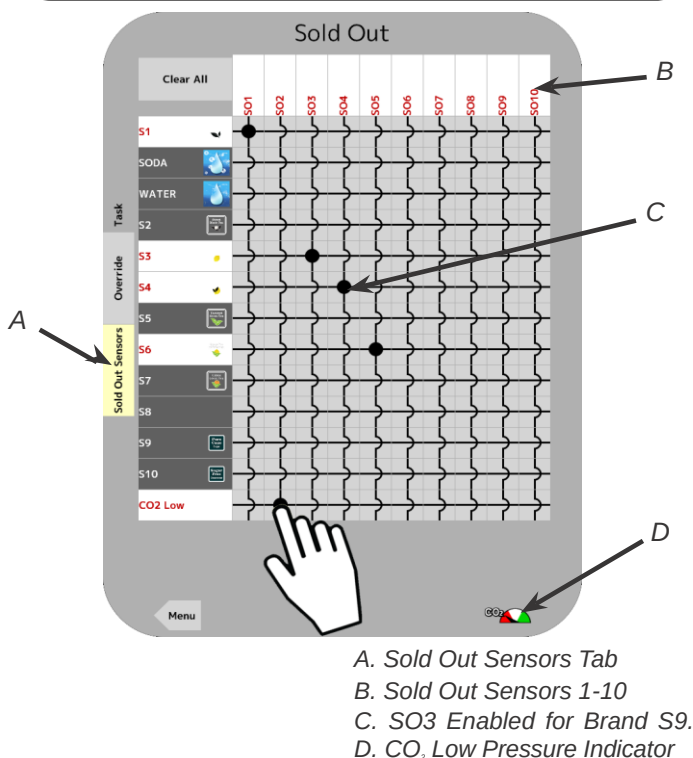


4. Press the *Sold Out Sensors* tab on the left side of the screen.

5. To add the Auto Sold Out feature to a specific brand, simply tap the corresponding intersection to activate a sold out sensor for that brand.

NOTE

This feature will automatically shut-off the pump for that specific brand when there is no product to be dispensed. This feature only comes into effect when the corresponding brand is changed to "Auto" in the Sold-Out menu.



- A. Sold Out Sensors Tab
- B. Sold Out Sensors 1-10
- C. S03 Enabled for Brand S9.
- D. CO₂ Low Pressure Indicator

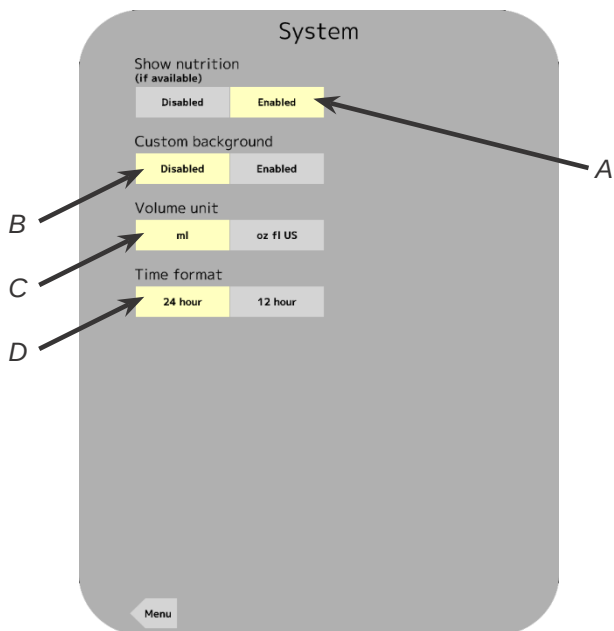
NOTE

If a Sold Out Sensor is utilized for the CO₂ low section, as shown in the image above, then the CO₂ Low Pressure Indicator will appear whenever the unit or a valve is not receiving enough CO₂.

6. Press the *Menu* button to return to the Service Menu.

Laitteen asetukset

1. Service menussa, paina *System* vasemmalta.
2. Paina *Enabled* nappia (A) jos haluat näyttää tuotteen ravintoarvot (ei käytetä).



- A. Ravintoarvot
- B. Taustakuva
- C. Mittayksiköt
- D. Aikaformaatti

3. Paina *Menu* palataksesi Service Menuun ja paina *Save Settings ja Exit*.

4. Press and hold the new *Nutrition Facts* “page curl” at the top of the screen to reveal the nutrition information of the brands/ flavors present on the screen (if available).

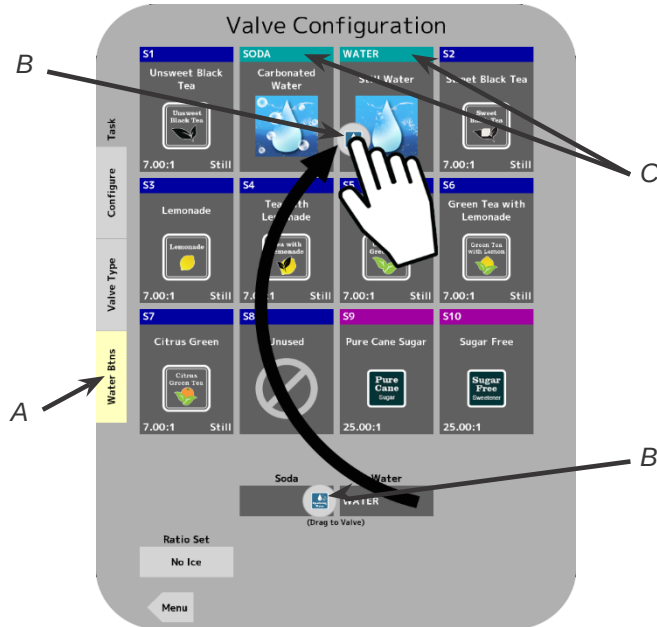


- A. Nutrition Page Curl
- B. Pour Button

5. *Custom background* vaihtaa taustakuvanjonka voi ladata muistitikulta (sivu 21).
7. *Volume unit* vaihtaa mittayksiköt
8. *Time format* vaihtaa 12h ja 24h kellon välillä

Water Button Feature

1. Jotta vettä ja soodaa voidaan laskea, paina Configuration Service menussa
2. Paina Water Btns välilehteä vasemmalla.
3. Täältä vesinapit (Sooda ja tavallinen) voidaan siirtää ja sijoittaa niiden moduuleihin.



- A. Water Buttons välilehti
B. Normi / soda nappi
C. Normi / Sooda moduuli

4. Kun vesinappi on raahattu oman moduulinsa päälle, se ilmestyy päävalikkoon..



- Vesinappi
päävalikossa

NOTE

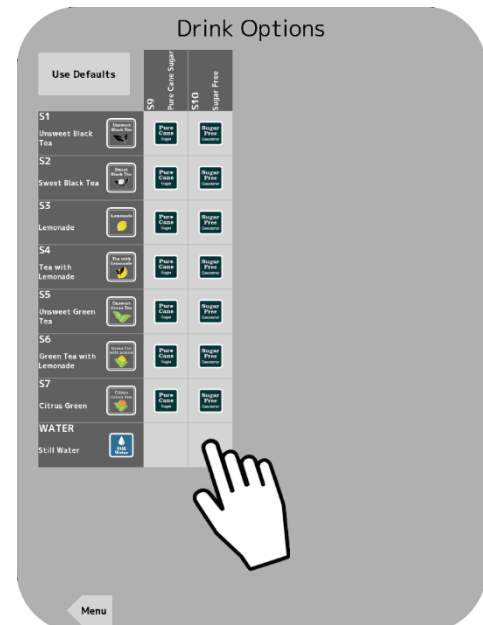
The carbonated/plain water valve must be present in order for water button to properly function. If there is no carb/plain water valve present, the corresponding water button will not appear on screen.

Drink Options Menu (Shotit ei käytössä)

1. From the Service Menu, press the Drink Options button to access the Drink Options Menu.
2. Tap next to any of the configured brands or water valves to enable or disable the use of any one of the configured flavor valves.

NOTE

When disabled, the designated flavor will not be selectable for that specific brand. It is recommended to disable flavor valves for both water buttons and any brand that is already sweetened.



DATAMANAGEMENT

Tuote päivitykset

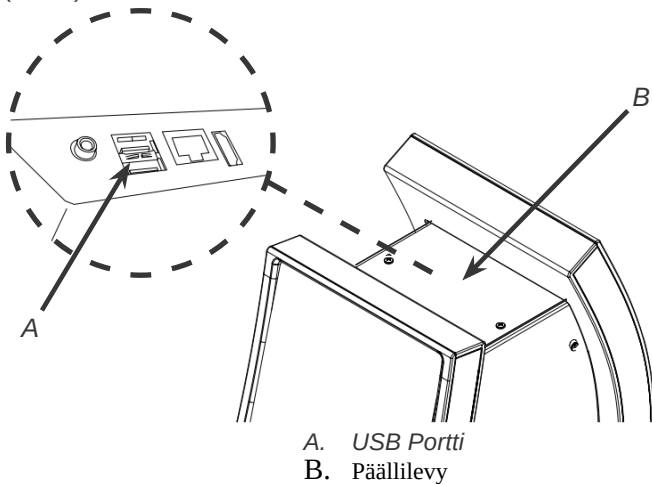
1. Using the *Tower Brand Management Software*, create the necessary *.brand* file, complete with new brand name and images.

IMPORTANT

Any data imported to the TsT will completely replace all existing content. For example: If the user wishes to add new brands to the existing brands library, the user must upload both the existing brand file as well as the new brand file(s).

For information on the Tower Brand Management Software visit lancercorp.com, contact your Lancer Customer Service Representative. Visit lancercorp.com and access the Tower Brand Management Software Instruction Sheet (Lancer Part Number: 28-2855/01).

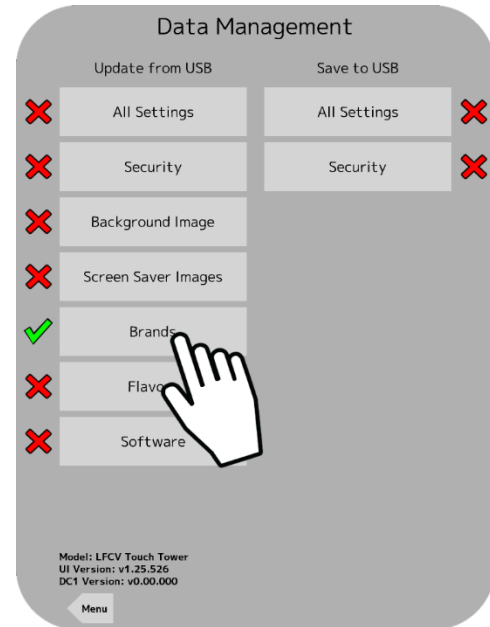
2. Luo tyhjälle muistitikulle Brands niminen kansio ja kopioi sinne *.brand*- loppuiset tiedostot.
3. Aseta tikku USB-porttiin joka on laitteen päällilevyn alla (sivu 8).



4. Service Menussa, paina *Data Management*.
5. "Update from USB" puolella, paina *Brands* nappia.

NOTE

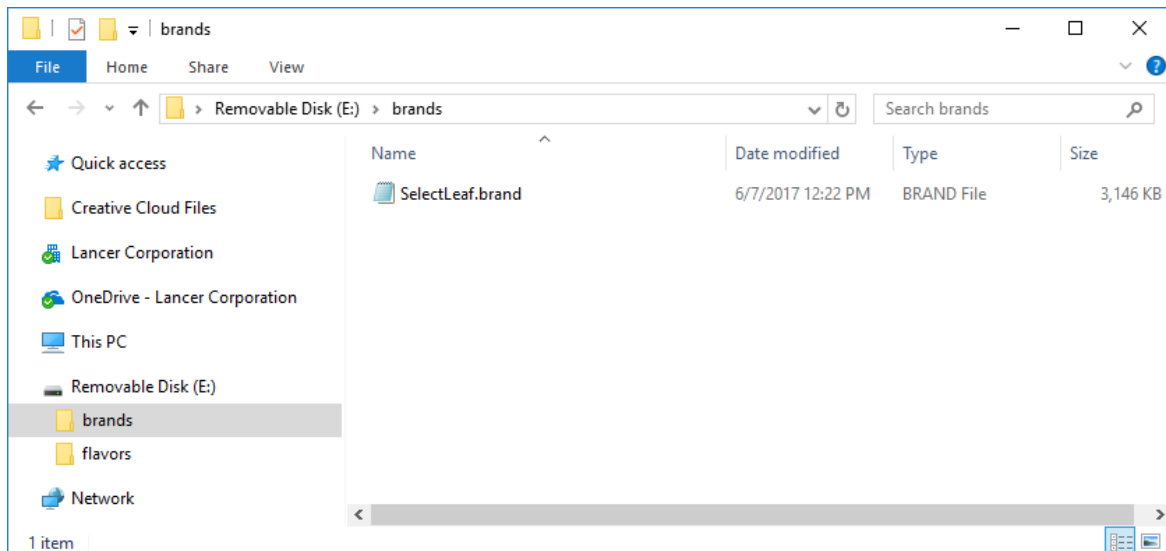
Brands laatikon vieressä on vihreä väkänen kun tiedostot ovat oikeassa paikassa



6. Kun Brands muuttuu vihreäksi, päivitykset ovat saatavilla.

NOTE

To upload new flavors to the TsT User Interface, create the flavor *.brand* file and put into a folder named "flavors", then repeat steps 2-5.



Näytönsäästäjän päivitys

1. Luo tikulle kansio Images ja sen sisälle kansio SS.
Kopioi näytönsäästäjävideo tänne

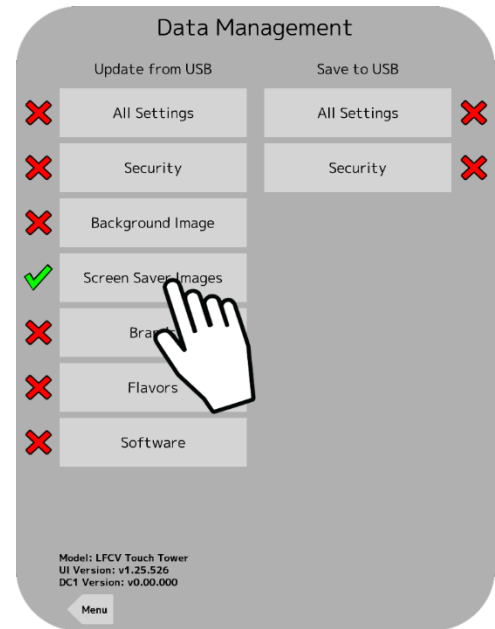
NOTE

Videon tulee olla .mp4 muodossa.
Resoluution tulee olla 768 px x 1024 px.

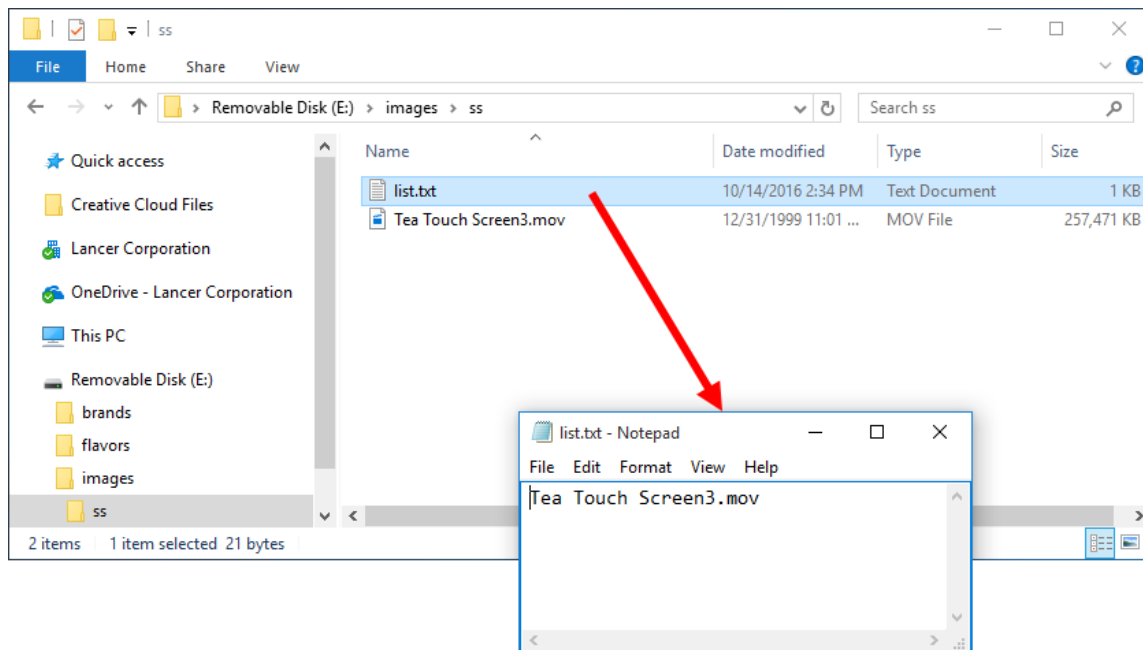
2. Luo .txt tiedosto jonne kirjoitat videon nimen.
Nimeä tiedosto nimellä: list.txt
3. Kun video ja .txt ovat SS- kansiossa, liitä tikku laitteeseen.

NOTE

Useita videoita voidaan ladata samallakertaa.
Lisää vain videoiden nimet .txt- tiedostoon.



5. Kun Screen Saver Images nappi muuttuu vihreäksi, käynnistä laite uudestaan ja uudet tiedostot ovat käytettävissä.



Taustakuvan päivitys

1. Luo tikulle uusi kansio nimeltä "images".

NOTE

The image file must be in a .png format and the dimensions of the video must be 768 px X 1024 px.

2. Vaihda kuvan nimeksi "background.png".
3. Kun *background.png* on "images" kansiossa, kytke tikku laitteeseen.

4. Service Menussa Data Managementissa, paina *Background Image* nappia.

NOTE

There will be a check mark next to the *Background Image* button if the USB drive has the image file in the correct place.

5. Kun *Background Image* nappi muuttuu vihreäksi, käynnistä laite uudestaan ja uudet tiedostot ovat käytettävissä.

Export Tower Settings

NOTE

The following highlights the steps necessary in order to copy a tower's brand configuration settings from one Tower dispenser to another.

1. Plug an empty USB Drive into the Tower port located underneath the head of the tower.
2. Access the Service Menu and under the Save to USB section, press the *All Settings* button.

NOTE

There will be a check mark next to the *All Settings* button if an empty USB drive is inserted.

3. Once the *All Settings* button turns green, remove the USB and plug into a different Tower.
4. Access the Service Menu and under the Update from USB section, press the *All Settings* button.

CLEANING AND SANITIZING

General Information

- Lancer equipment (new or reconditioned) is shipped from the factory cleaned and sanitized in accordance with NSF guidelines. The operator of the equipment must provide continuous maintenance as required by this manual and/or state and local health department guidelines to ensure proper operation and sanitation requirements are maintained.

NOTE

The cleaning procedures provided herein pertain to the Lancer equipment identified by this manual. If other equipment is being cleaned, follow the guidelines established by the manufacturer for that equipment.

- Cleaning should be accomplished only by trained personnel. Sanitary gloves are to be used during cleaning operations. Applicable safety precautions must be observed. Instruction warnings on the product being used must be followed.

! ATTENTION

- Use sanitary gloves when cleaning the unit and observe all applicable safety precautions.
- **DO NOT** use a water jet to clean or sanitize the unit.
- **DO NOT** disconnect water lines when cleaning and sanitizing syrup lines, to avoid contamination.
- **DO NOT** use strong bleaches or detergents; These can discolor and corrode various materials.
- **DO NOT** use metal scrapers, sharp objects, steel wool, scouring pads, abrasives, or solvents on the dispenser.
- **DO NOT** use hot water above 140° F (60° C). This can damage the dispenser.
- **DO NOT** spill sanitizing solution on any circuit boards. Insure all sanitizing solution is removed from the system.

Cleaning and Sanitizing Solutions

Cleaning Solution

Mix a mild, non-abrasive detergent (e.g. Sodium Laureth Sulfate, dish soap) with clean, potable water at a temperature of 90°F to 110°F (32°C to 43°C). The mixture ratio is one ounce of cleaner to two gallons of water. Prepare a minimum of five gallons of cleaning solution. Do not use abrasive cleaners or solvents because they can cause permanent damage to the unit. Ensure rinsing is thorough, using clean, potable water at a temperature of 90°F to 110°F. Extended lengths of syrup lines may require additional cleaning solution.

! CAUTION

If a powder sanitizer is used, dissolve it thoroughly with hot water prior to adding to the product system. Ensure sanitizing solution is removed from the dispenser as instructed.

Sanitizing Solution

Prepare sanitizing solutions in accordance with the manufacturer's written recommendations and safety guidelines. The solution must provide 100 parts per million (PPM) chlorine (e.g. Sodium Hypochlorite or bleach). A minimum of five gallons of sanitizing solution should be prepared. Any sanitizing solution may be used as long as it is prepared in accordance with the manufacturer's written recommendations and safety guidelines, and provides 100 parts per million (PPM) chlorine.

Cleaning and Sanitizing Syrup Lines

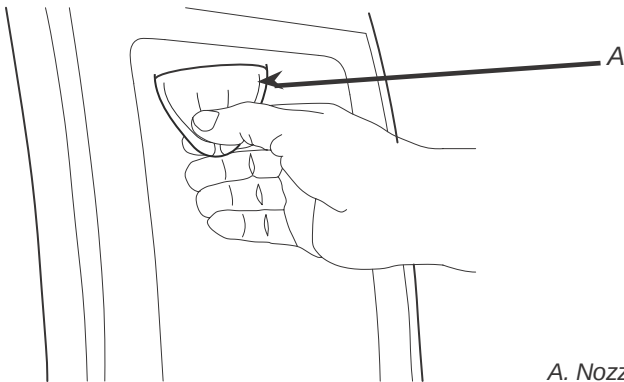
1. Disconnect syrup lines from BIB's or other product supply.
2. Place syrup lines, with BIB connectors, in a bucket of warm water.
3. Activate each valve to fill the lines with warm water and flush out syrup remaining in the lines.
4. Prepare Cleaning Solution described above.
5. Place syrup lines, with BIB connectors, into cleaning solution.
6. Activate each valve until lines are filled with cleaning solution then let stand for ten (10) minutes.
7. Flush out cleaning solution from the syrup lines using clean, warm water.
8. Prepare Sanitizing Solution described above.
9. Place syrup lines into sanitizing solution and activate each valve to fill lines with sanitizer. Let sit for ten (10) minutes.
10. Reconnect syrup lines to BIB's and draw drinks to flush solution from the dispenser.
11. Taste the drink to verify that there is no off-taste. If off-taste is found, flush syrup system again.

! CAUTION

Following sanitization, rinse with end-use product until there is no aftertaste. Do not use a fresh water rinse. This is a NSF requirement. Residual sanitizing solution left in the system creates a health hazard.

Cleaning and Sanitizing Nozzle

1. Disconnect power, so as to not activate valve while cleaning.
2. Remove outer nozzle by twisting counter clockwise and pulling down.
3. Rinse nozzle with warm water.
4. Wash nozzle with cleaning solution then immerse in sanitizing solution and let sit for fifteen (15) minutes.
5. Set nozzle aside and let air dry. **DO NOT** rinse with water after sanitizing.
6. Using a soft, clean cloth and cleaning solution, clean the nozzle injectors.
7. Using a soft, clean cloth, sanitize the nozzle injectors and let air dry.
8. Reconnect nozzle.
9. Connect power.
10. Taste the drink to verify that there is no off-taste. If off-taste is found, sanitize the nozzle and nozzle injectors again.

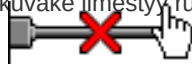


A. Nozzle

! CAUTION

Following sanitization, rinse with end-use product until there is no aftertaste. Do not use a fresh water rinse. This is a NSF requirement. Residual sanitizing solution left in the system creates a health hazard.

Vianhaku

Vika	Syy	Ratkaisu
Asennuksen jälkeen näyttö ei reagoi kosketukseen	1. Näyttö on likainen	1. Puhdista kuivalla liinalla
Tämä kuvake ilmestyy ruutuun:	1. Yhteysongelma	1. Tarkista yhteys laitteen ja näytön välillä
Tämä kuvake ilmestyy ruutuun : 	1. Kosketusnäytön yhteys on katkennut	1. Ota yhteys Lancerin asiakaspalveluun
Vesivuoto suuttimella.	1. Viallinen tai puuttuva O-rengas.	1. Vaihda O-rengas.
Muu vuoto.	1. Rakoa liittimissä tms.. 2. Vaurioitunut tai väärin asennettu O-rengas	1. Kiristä liitokset 2. Vaihda tai kiristä O-re
Insufficient water flow.	1. Insufficient incoming supply water pressure. 2. Shutoff on mounting block not fully open. 3. Foreign debris in water flow control. 4. Foreign debris in water pump strainer	1. Verify incoming supply water pressure is a minimum of 25 PSI (0.172 MPa). 2. Open shutoff fully. 3. Remove water flow control from upper body and clean out any foreign material to ensure smooth free spool movement. 4. Remove water pump strainer and clean.
Insufficient syrup flow.	1. Insufficient CO ₂ pressure to BIB pumps. 2. Out of CO ₂ . 3. Shutoff on mounting block not fully open. 4. Foreign debris in syrup flow control. 5. Bad syrup pump.	1. Adjust CO ₂ pressure to 80 PSI (0.550 MPa) [minimum 70 PSI (0.480 MPa)] for BIB pumps. 2. Replace CO ₂ tank/refill. 3. Open shutoff fully. 4. Remove syrup flow control from upper body and clean out any foreign material to ensure smooth free spool movement. 5. Replace BIB pump.
Erratic ratio.	1. Incoming water and/or syrup supply not at minimum flowing pressure. 2. Foreign debris in water and/or syrup flow controls.	1. Check pressure and adjust 2. Remove flow controls from upper body and clean out any foreign material to ensure smooth free spool movement.
No syrup dispensed	1. Water and syrup shutoffs on mounting block not fully open. 2. Electric current not reaching valve. 3. Improper or inadequate water or syrup supply. 4. Transformer Failure 5. Bad valve solenoid(s)	1. Open shutoff fully. 2. Check electric current supplied to valve. If current is adequate, check solenoid coil and switch, and replace if necessary. 3. Remove valve from mounting block and open shutoffs slightly and check water and syrup flow. If no flow, check dispenser for freeze-up or other problems 4. Reset transformer circuit breaker. If breaker trips again check for pinched wire harness at back blocks 5. Replace Solenoid(s)

TROUBLE	CAUSE	REMEDY
Water only dispensed; no syrup; or syrup only dispensed, no water	1. Water or syrup shutoff on mounting block not fully open. 2. Improper or inadequate water or syrup flow. 3. BIB supply too far from dispenser. 4. CO₂ pressure too low. 5. Stalled or inoperative BIB pump 6. Kinked line.	1. Open shutoff fully. 2. Remove valve from mounting block, open shutoffs slightly and check water and syrup flow. If no flow, check dispenser for freeze-up or other problems. Ensure BIB connection is engaged. 3. Check that BIB supply is within six (6) feet of the dispenser. 4. Check the CO₂ pressure to the pump manifold to ensure it is between 70 and 80 psi (0.483 and 0.552 MPa). 5. Check CO₂ pressure and/or replace pump. 6. Remove kink or replace line.
Syrup only dispensed. No water, but CO ₂ gas dispensed with syrup.	1. Improper water flow to dispenser. 2. Carbonator pump motor has timed out.	1. Check for water flow to dispenser (see Insufficient Water Flow on previous page). 2. Reset by turning the unit OFF and then ON.
Excessive foaming.	1. Incoming water or syrup temperature too high. 2. CO₂ pressure too high. 3. Water flow rate too high. 4. Nozzle not installed correctly. 5. Nozzle and nozzle injectors not clean. 6. Air in BIB lines. 7. Poor quality ice. 8. High beverage temperature.	1. Correct prior to dispenser. Consider larger dispenser or pre-cooler. 2. Adjust CO₂ pressure downward, but not less than 70 PSI (0.483 MPa). 3. Re-adjust and reset ratio. Refer to "Adjust Water Flow Rate & Syrup/Water Ratio" Section. 4. Remove and reinstall properly. 5. Remove nozzle and clean injectors. 6. Bleed air from BIB lines. 7. Check quality of ice used in drink. 8. Check refrigeration system.
Warm drinks.	1. Restricted airflow. 2. Dispenser connected to hot water supply. 3. Dispenser capacity exceeded.	1. Check clearances around sides, top, and inlet of unit. Remove objects blocking airflow through grill. 2. Switch to cold water supply. 3. Add pre-cooler or replace with larger dispenser.
Circuit breaker tripping.	1. Valve wire harness shorted to itself or to faucet plate. 2. PCB is bad. 3. Secondary wire harness is bad. 4. Transformer failure.	1. Detect short by disconnecting input fasten to key lock and single pin connector. Restore power if breaker doesn't trip. Then valve wire harness is shorted. If OK, reconnect. 2. Detect short by disconnecting J1 connector (24 VAC input) from PCB. Restore power, if breaker doesn't trip. Then replace PCB. If breaker does trip, then PCB is OK. Reconnect J1 connector. 3. If it does not trip, locate short in secondary harness between transformer, PCB, and valve wire harness. 4. Detect short by disconnecting both transformer fastons and restore power. If breaker does trip, replace transformer.

TROUBLE	CAUSE	REMEDY
BIB pump does not operate when dispensing valve opened.	1. Out of CO₂, CO₂ not turned on, or low CO₂ pressure. 2. Out of syrup. 3. BIB connector not tight. 4. Kinks in syrup or gas lines. 5. Bad BIB Pumps.	1. Replace CO₂ supply, turn on CO₂ supply, or adjust CO₂ pressure to 70-80 psi (0.483-0.552 MPa) 2. Replace syrup supply. 3. Fasten connector tightly. 4. Straighten or replace lines. 5. Replace BIB pump.
BIB pump operated, but no flow.	1. Leak in syrup inlet or outlet line. 2. Defective BIB pump check valve.	1. Replace line. 2. Replace BIB pump
BIB pump continues to operate when bag is empty.	1. Leak in suction line. 2. Leaking o-ring on pump inlet fitting.	1. Replace line. 2. Replace o-ring.
BIB pump fails to restart after bag replacement.	1. BIB connector not on tight. 2. BIB connector is stopped up. 3. Kinks in syrup line 4. Bad BIB Pumps.	1. Tighten BIB connector. 2. Clean out or replace BIB connector. 3. Straighten or replace line. 4. Replace BIB pump.
BIB pump fails to restart when dispensing valve is closed.	1. Leak in discharge line or fittings. 2. Empty BIB. 3. Air leak on inlet line or bag connector.	1. Repair or replace discharge 2. Replace BIB. 3. Repair or replace.
Low or no carbonation.	1. Low or no CO₂. 2. Excessive water pressure. 3. Worn or defective carbonator pump.	1. Check CO₂ supply. Adjust CO₂ pressure to 70 psi (0.483 MPa). 2. Water regulator should be set at 65 psi (0.448 MPa) 3. Replace carbonator pump.

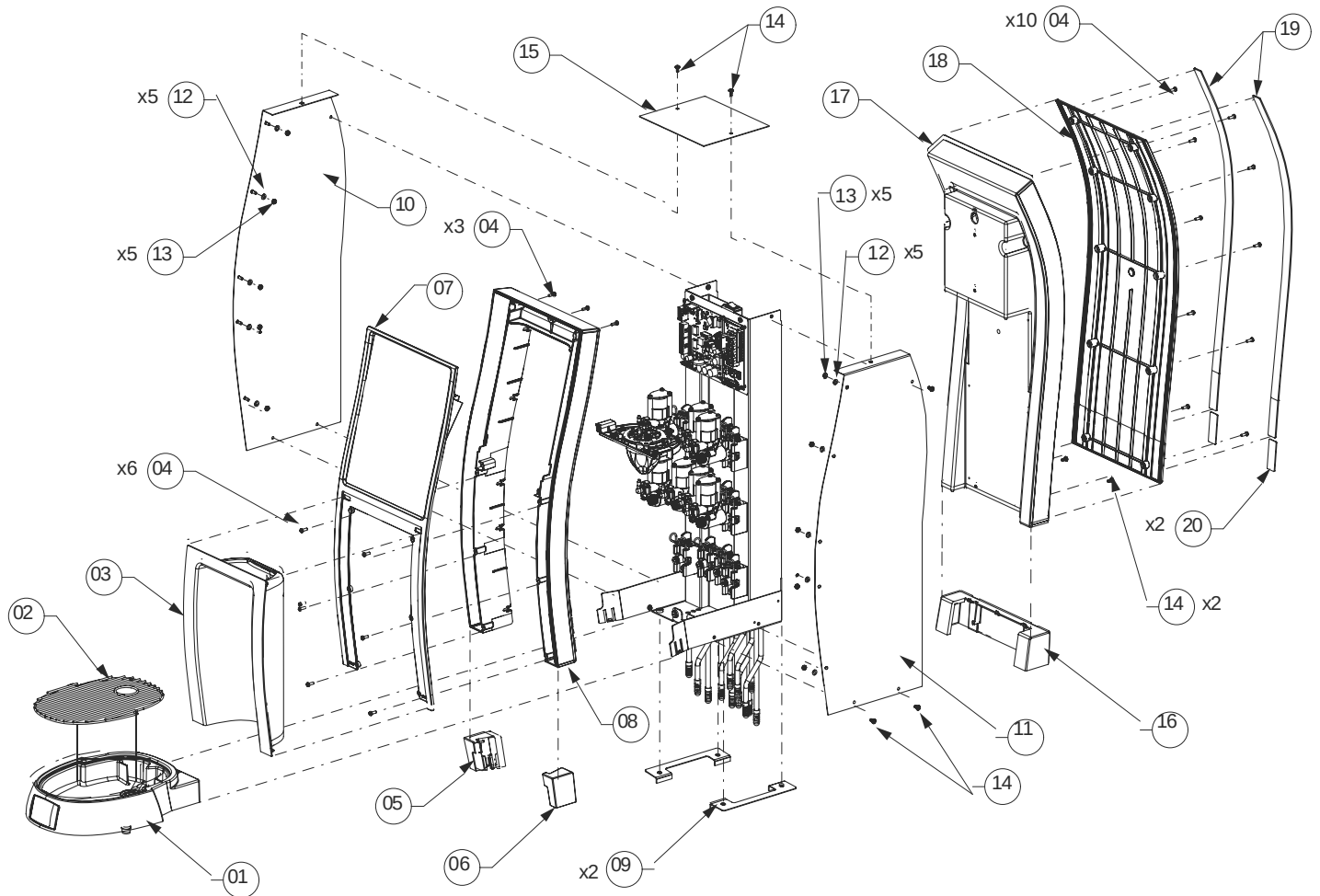
Dispenser Disposal



To prevent possible harm to the environment from improper disposal, recycle the unit by locating an authorized recycler or contact the retailer where the product was purchased. Comply with local regulations regarding disposal of the refrigerant and insulation.

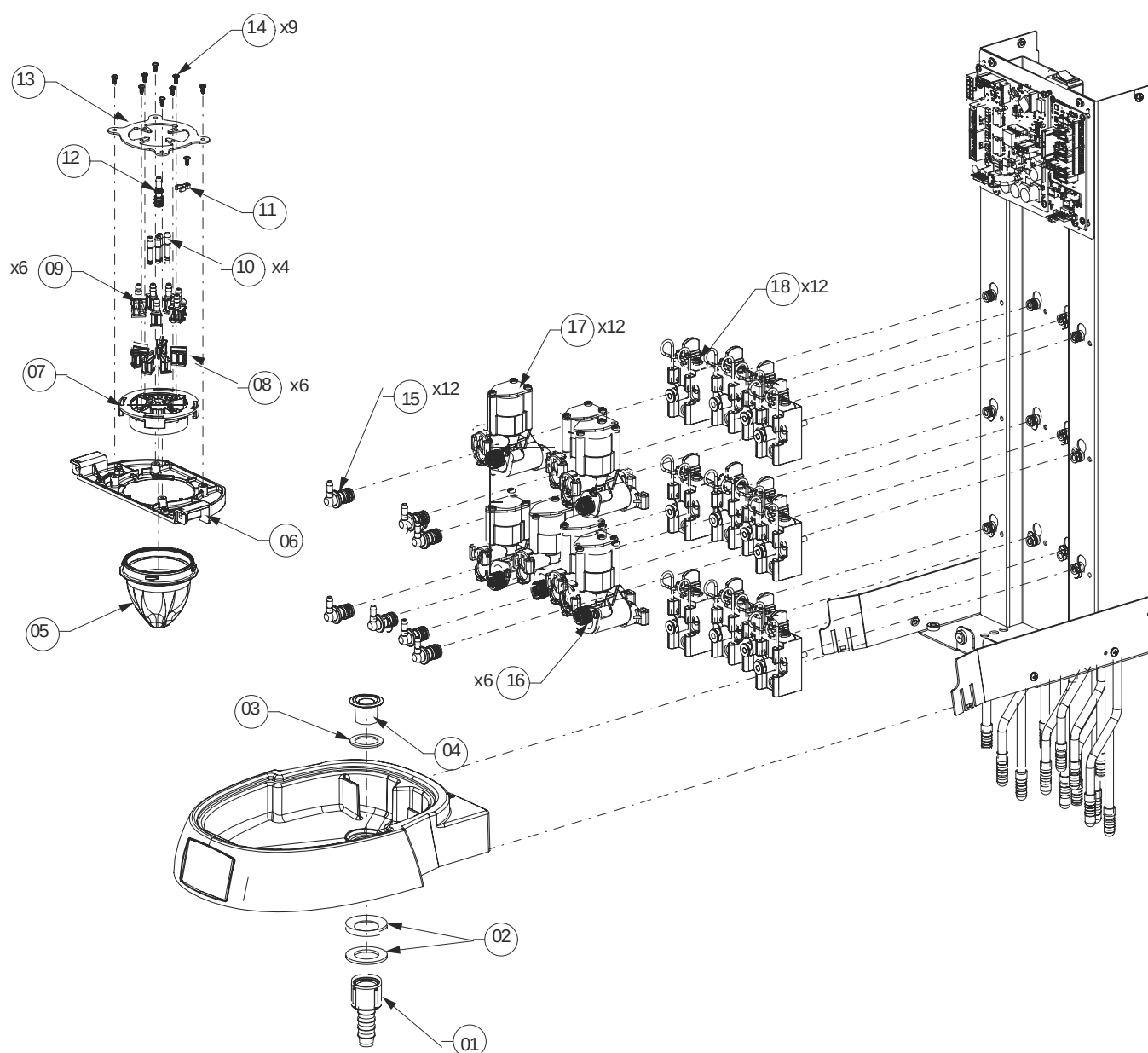
ILLUSTRATIONS AND PART LISTINGS

Main Unit Assembly



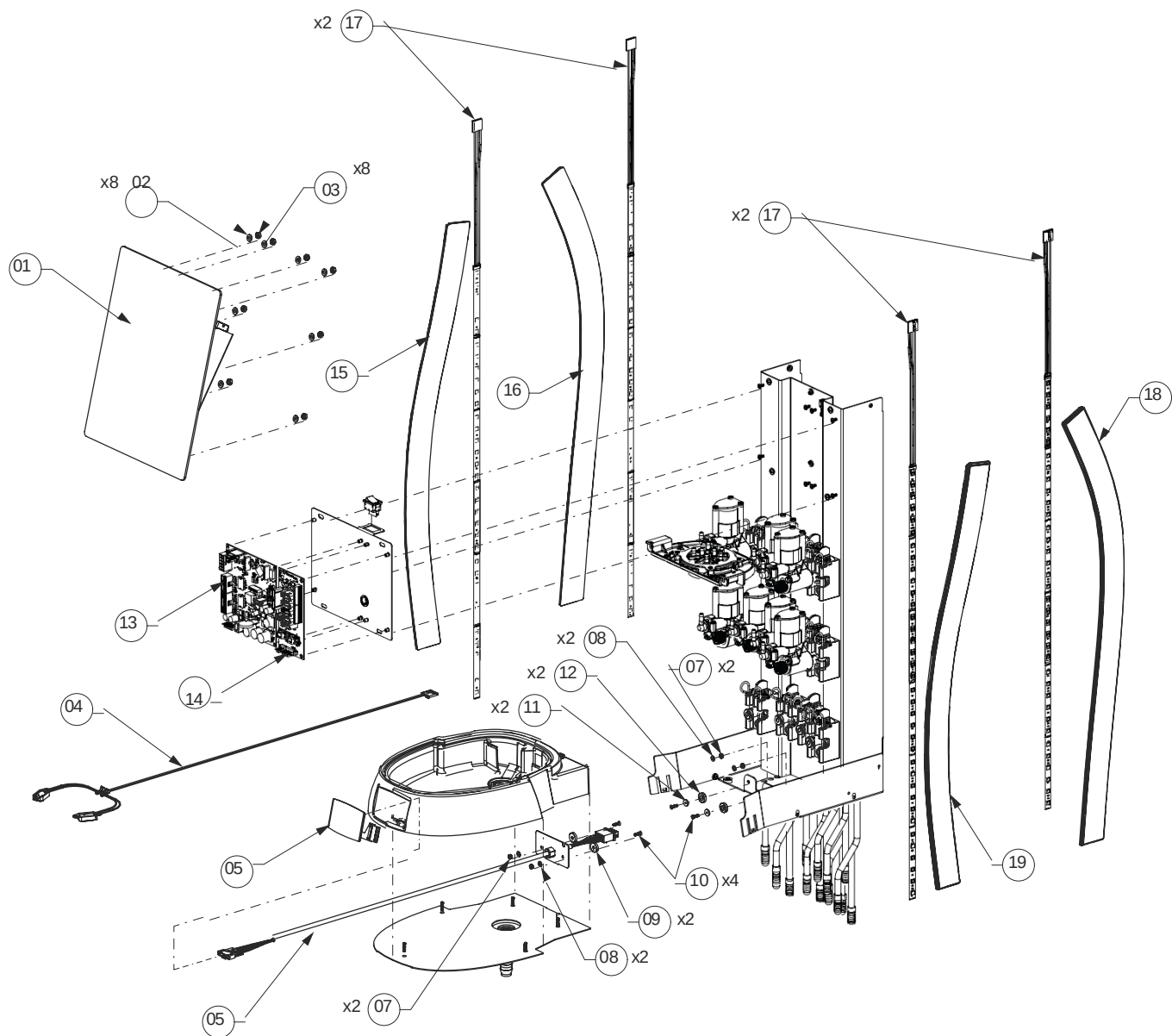
<u>Item</u>	<u>Part No.</u>	<u>Description</u>			
01	05-3429/01	Drip Tray, TsT	11	30-11889	Center Panel, Right
02	23-1726	Cup Rest, TsT	12	04-1714	Washer, M4
03	05-3296/01	Splash Plate, TsT	13	04-1717	Nut, M4
04	04-0633/01	Screw, 6-19 x 0.437	14	04-1689/01	Screw, M4
05	05-3434/01	Drip Tray Bumper, Left	15	30-11890	Center Panel, Top
06	05-3432/01	Drip Tray Bumper, Right	16	05-3437	Lower Rear Cover
07	05-3295/01	Front Cosmetic Panel	17	05-3292/01	Rear Internal Panel
08	05-3294/01	Front Internal Panel	18	05-3293/01	Rear Cosmetic Panel
09	30-16121	Mounting Plate Bracket, TsT	19	05-3435	Upper Rear Trim
10	30-11888	Center Panel, Left	20	05-3436	Lower Rear Trim

Plumbing Assembly



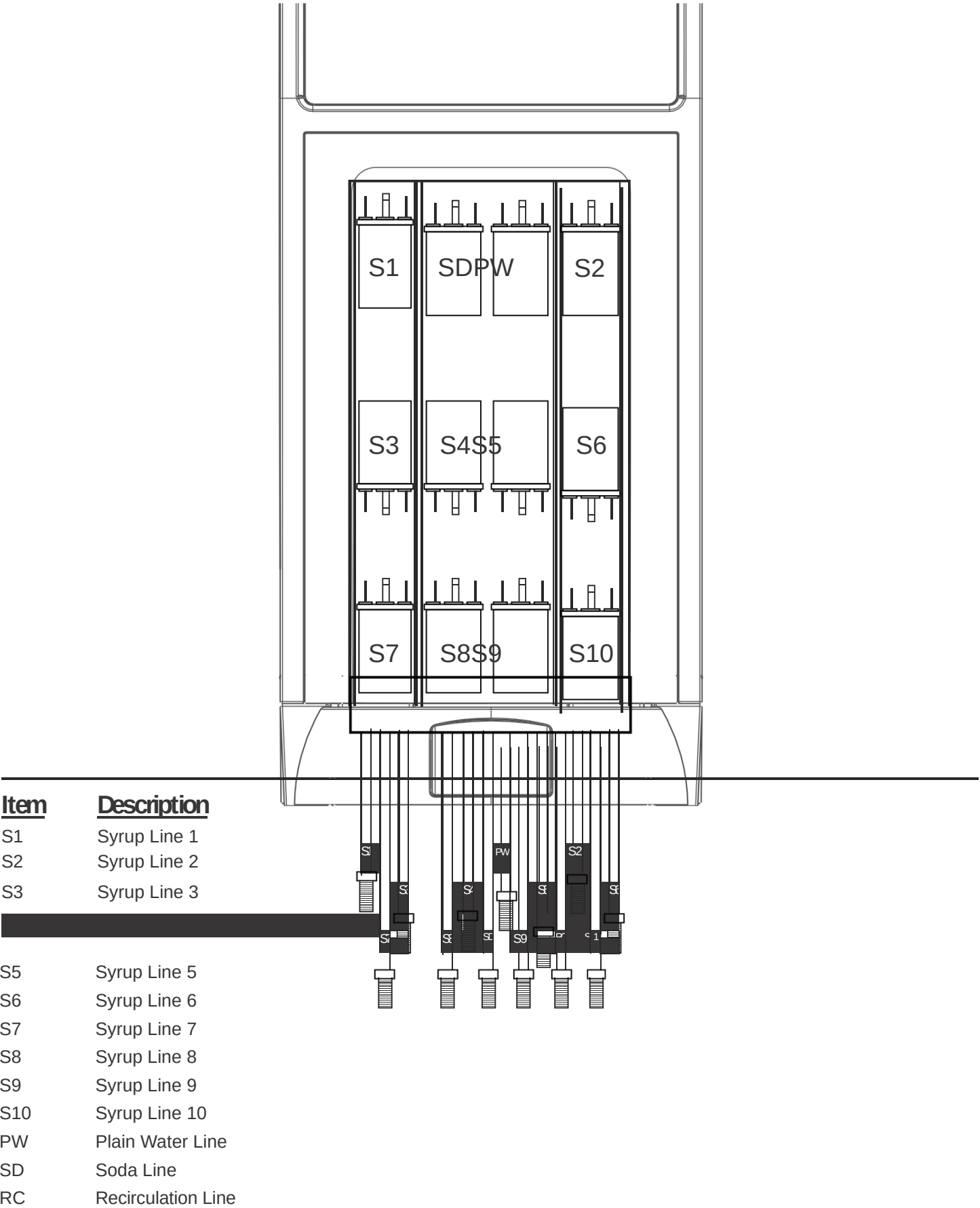
Item	Part No.	Description			
01	01-2991	Coupling, Female	10	05-3392	Injector, Blank, Bonus Flavor
02	04-1710/01	Washer, Flat	11	05-3309	Locking Clip, Water Inlet
03	02-0677	Seal, Washer, Drain	12	05-1612	Fitting, Water, M/F Nozzle
04	01-2994	Drain Fitting, 1/2 NPT	13	30-12139/01	Nozzle Plate
05	05-3407/01	Nozzle Overmold Assy	14	04-1693/01	Screw, 4-20 X 0.250
06	05-3297	Nozzle Mount, TsT	15	05-1385	Elbow, .5 Dole
07	05-3286/01	Inner Body, Flex Nozzle	16	19-0266/02	Valve Assy, 2.0 Syrup Blk
08	05-3391/01	Injector, Blank, Syrup	17	19-0267/02	Valve Assy, 2.0 Syrup Gray
09	05-3393/01	Injector, Syrup, Low Flow	18	82-2317/01	Mounting Block Assy

Electronics Assembly

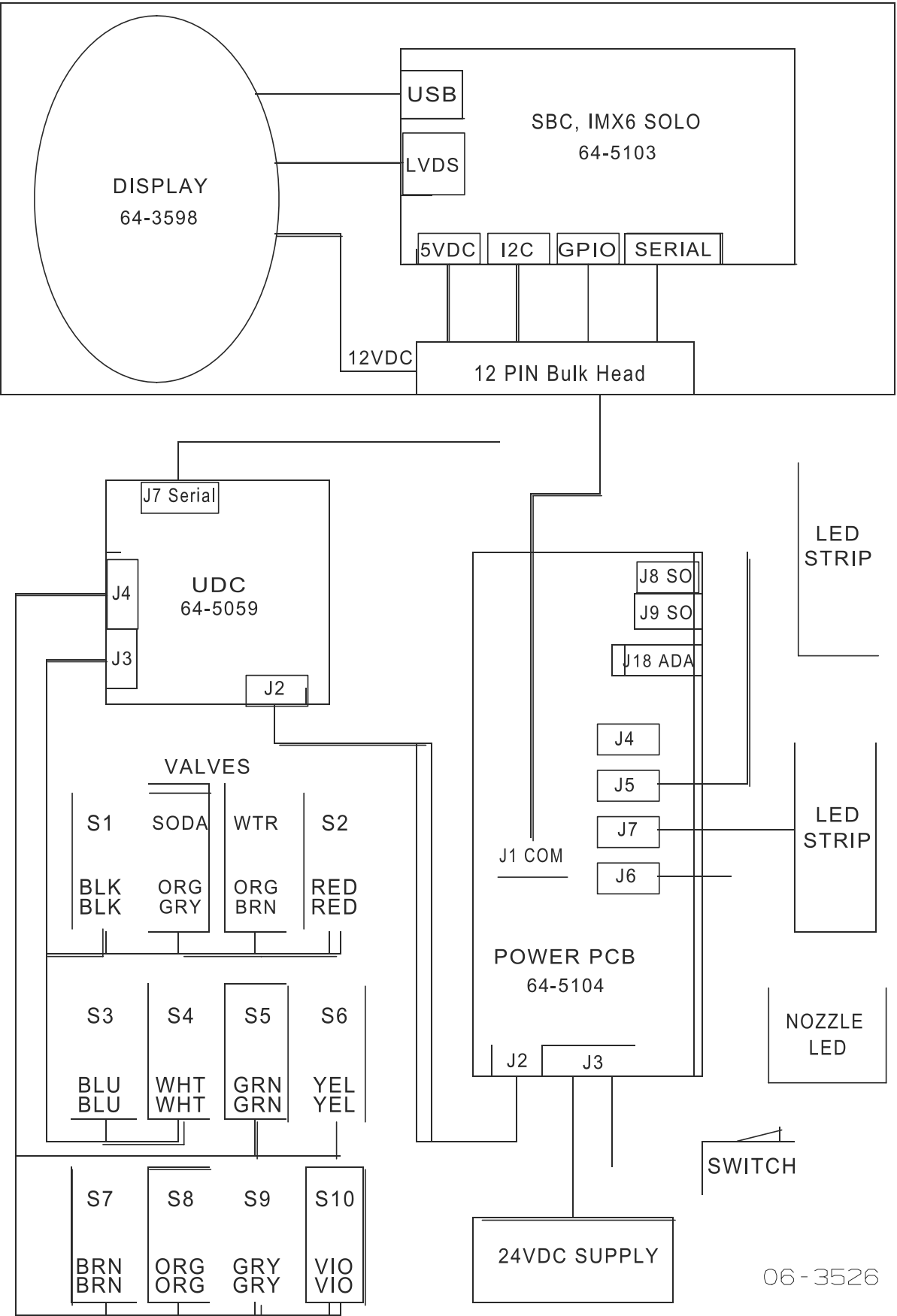


Item	Part No.	Description
01	82-4963/01	Screen Assy, TsT
02	04-1714	Washer, Flat, M4
03	04-1717	Nut, M4
04	52-3661	Drip Tray Dual Light Assy
05	12-0629	Keypad, Membrane Switch
06	52-3718	Harness, Keypad to ADA
07	04-1709/01	Hex Nut, M3 x 0.5, SS
08	04-1716	Washer, Flat, M3
09	10-0958	Magnet, Neo, 0.5 X 0.125
10	04-1748	Screw, M3
11	02-0681	Washer, 0.343 OD X 0.093
12	10-1000	Magnet, Neo, Reverse, 0.5 X 0.125
13	64-5061	PCB Assy, Valve Board
14	64-5104/01	PCB Assy, Power Board
15	05-3483	Front Lens, Left
16	05-3486	Rear Lens, Left
17	52-3730	LED Strip Assy
18	05-3485	Rear Lens, Right
19	05-3484	Front Lens, Right

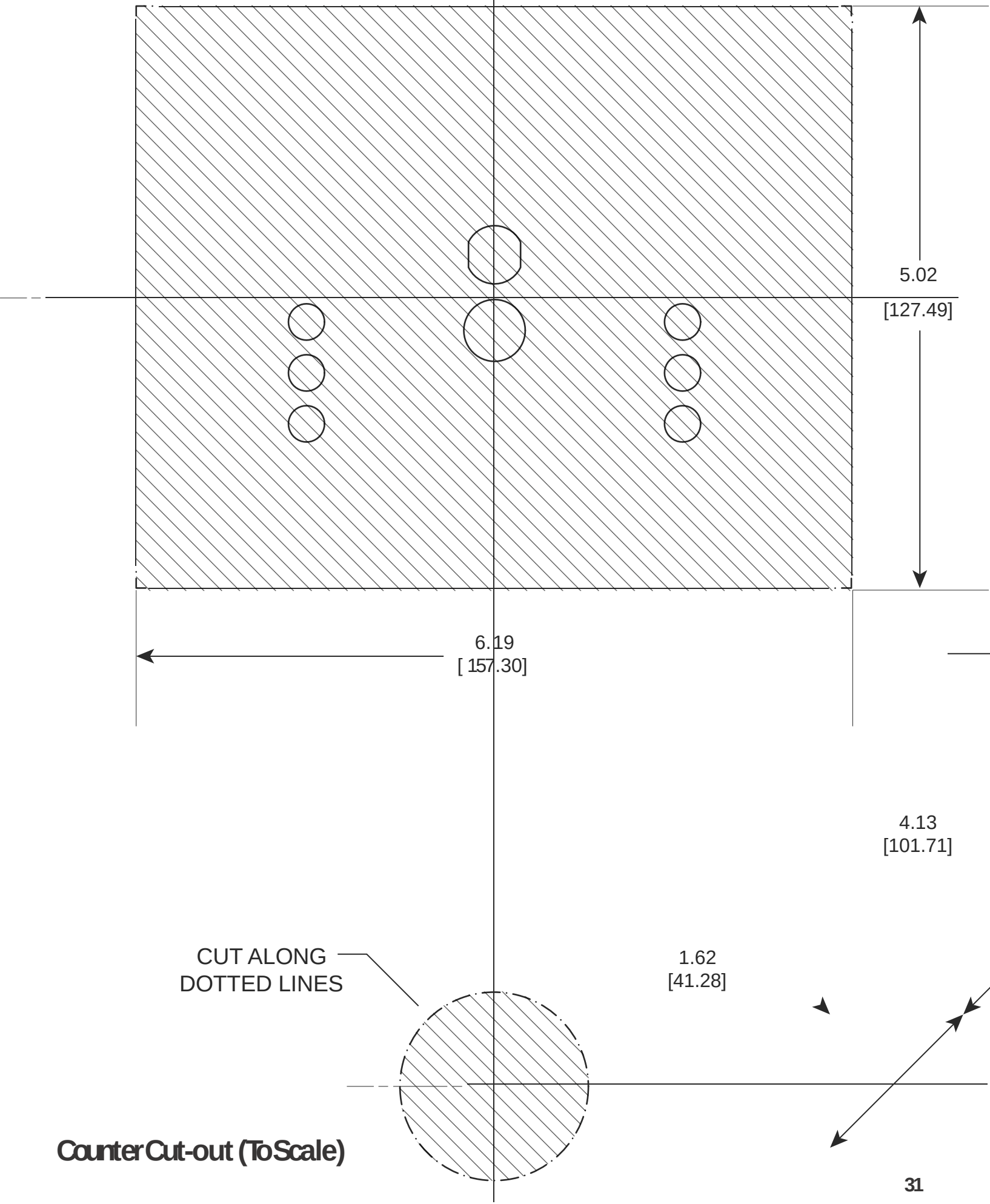
Unit Plumbing Diagram



Unit Wiring Diagram



06-3526



FRONT OF UNIT

LANCER[®]

Lancer Corp.

800-729-1500

Technical Support/Warranty: 800-729-1550

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lancercorp.com